

CHAPTER 01 - Influence Lines of Determinate and Indeterminate Structures

I. Influence Lines of Determinate Structures

I.1. Introduction

I.2. Influence lines for beams: Static–analytical approach

I.3. Müller–Breslau principle: kinematic–qualitative approach

I.4. Influence lines for Girders with floor systems

I.5. Influence lines for Trusses

I.6. Influence lines for Frames

I.7. Applications of influence lines

I.7.1. Response at a particular location due to a single moving concentrated load

I.7.2. Response at a particular location due to a uniformly distributed live load

I.7.3. Response at a particular location due to a series of moving concentrated loads

I.8. Problems

II. Influence Lines of Indeterminate Structures

Influence Lines of determinate and Indeterminate Structures

1. Introductoion

Influence lines have importante application for the design of structures that resist large live loads. In this module we will discuss how to draw the influence line for a statically determinate structure. The theory is applied to structures subjected to a distributed load or a series of concentrated forces, and specific applications to floor girders and bridge trusses are given. The determination of the absolute maximum live shear and moment in a member is discussed at the end of this module

2. Learning outcomes: apply the methods of structural analysis for external and internal actions on various types of structural systems due to static and moving loads.

3. Learning objectives

Student learning objectives for this module upon completion of this module, you will be able to:

- Know the meaning of influence lines.
- Determine its an important part in the design of bridges, industrial crane rails, conveyors, and other structures where loads move across their span.

4. Learning activities

Activities for this module

- Do the assignments and problem sets
- Take the quiz (Quiz1)

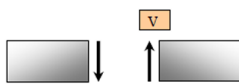
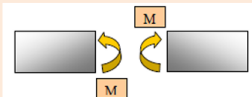
I. Influence Lines of determinate Structures

I.1. Defintion

An influence line represents the variation of the reaction, shear, moment, or deflection at a specific point in a member as a concentrated force moves over the member. Once this line is constructed, one can tell at a glance where the moving load should be placed on the structure so that it creates the greatest influence at the specified point. Furthermore, the magnitude of the associated reaction, shear, moment, or deflection at the point can then be calculated from the ordinates of the influence line diagram. For these reasons, influence lines play an important part in the design of bridges, industrial crane rails, conveyors, and other structures where loads move across their span.

The following sign conventions are used:

Table 1. sign conventions

Paramètre	Signe de la ligne d'influence
Réaction	Positif vers le haut
Effort tranchant	
Moment fléchissant	

 **Remark:** For other sign conventions, they must be clearly defined.

I.2 Influence Lines for Beams

I.2.1 Procedure for Analysis (Analytical Method)

Either of the following two procedures can be used to construct the influence line at a specific point P in a member for any function (reaction, shear, or moment). For both of these procedures we will choose the moving force to have a dimensionless magnitude of unity.

I.2.1 Tabulate Values

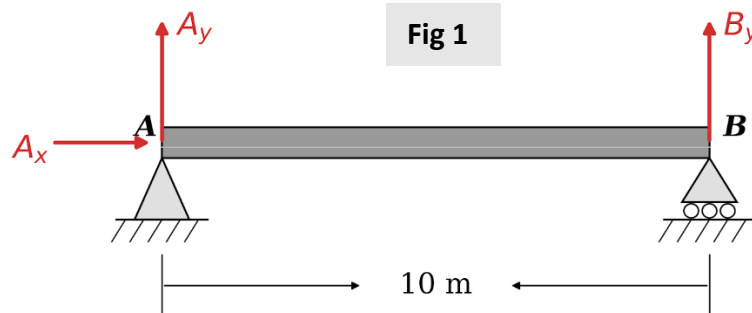
- Place a **unit load** $P = 1$ at different positions x along the member.
- For each position, use **statics** to calculate the required quantity: **reaction** R , **shear force** V , or **bending moment** M .
- Apply the adopted **sign conventions**.
- For a **statically determinate beam**, the influence line consists of **straight-line segments**.
- Prepare a **table** showing the unit-load position x and the corresponding value of R , V , or M .
- Plot the calculated values and **connect the points** to construct the influence line.

Key principle:

Unit load $P = 1 \rightarrow$ Calculate R , V , or $M \rightarrow$ Tabulate values $\rightarrow$ Plot the influence line

Example 01

Construct the influence line for the vertical reaction at A of the beam shown in Fig. 1 using the analytical method by tabulating the values.



SOLUTION

Tabulate Values. A unit load is placed on the beam at each selected point x and the value of A_y is calculated by summing moments about B. For example, when $x=2.5$ m, $x=5$ m and $x=7$ m, see **Figs. 2-3** and **4**, respectively. The results for A_y are entered in the **Table 2**. A plot of these values yields the influence line for the reaction at A, **Fig. 5**. This line or diagram gives the reaction at A as the unit load moves from one position to the next along the beam.

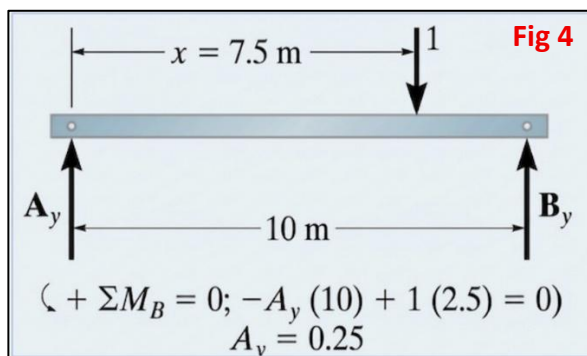
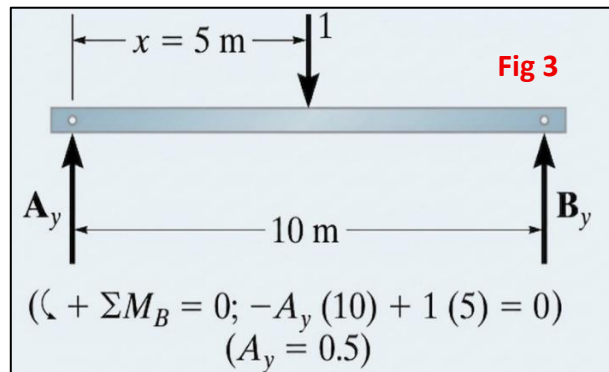
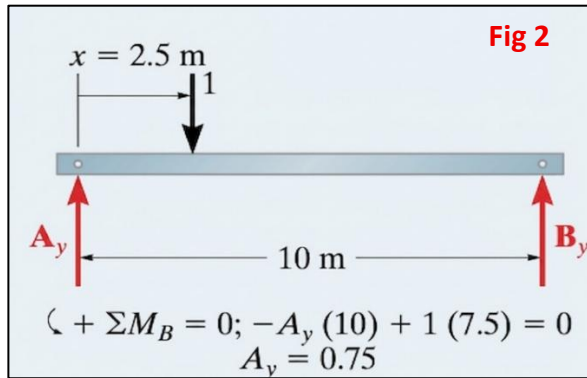
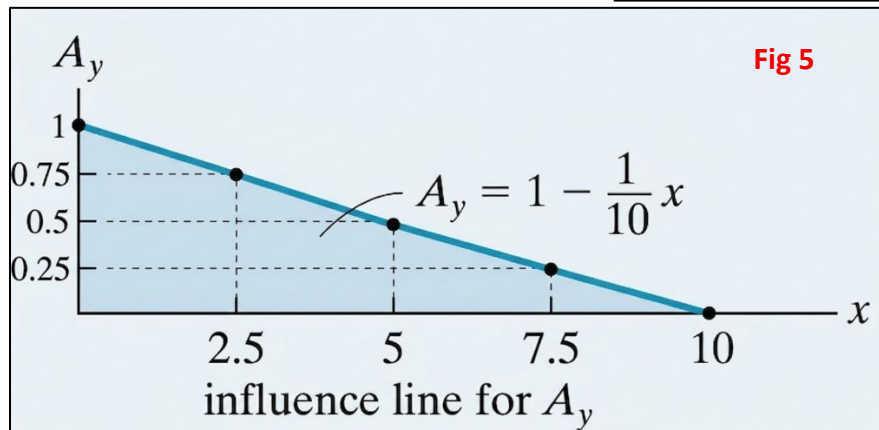


Table 2

x	A_y
0	1
2.5	0.75
5	0.5
7.5	0.25
10	0



1.2.2 Influence-Line Equations

Rather than placing the unit load at specific points on the beam and plotting the values of A_y for each point, we can place the unit load a variable distance x from A, **Fig 6**. The reaction A_y as a function of x can be determined from.

Influence Line Equation

The equation expressing the reaction A_y as a function of the position x of the unit load ($P = 1$) on the 10 m beam is:

$$A_y(x) = 1 - \frac{1}{10}x \quad \text{for } 0 \leq x \leq 10 \text{ m}$$

Step-by-Step Derivation Method

1. Definition of distances :

- The total span of the beam is $L = 10 \text{ m}$.
- The unit load $P = 1$ is located at a distance x from support A.

- The lever arm of the load P relative to support B is $(10 - x)$.

2. **Moment equilibrium equation about B ($\sum M_B = 0$):**

Taking counterclockwise as positive, the reaction A_y creates a clockwise (negative) moment and the unit load 1 creates a counterclockwise (positive) moment about B :

$$-A_y \cdot (10) + 1 \cdot (10 - x) = 0$$

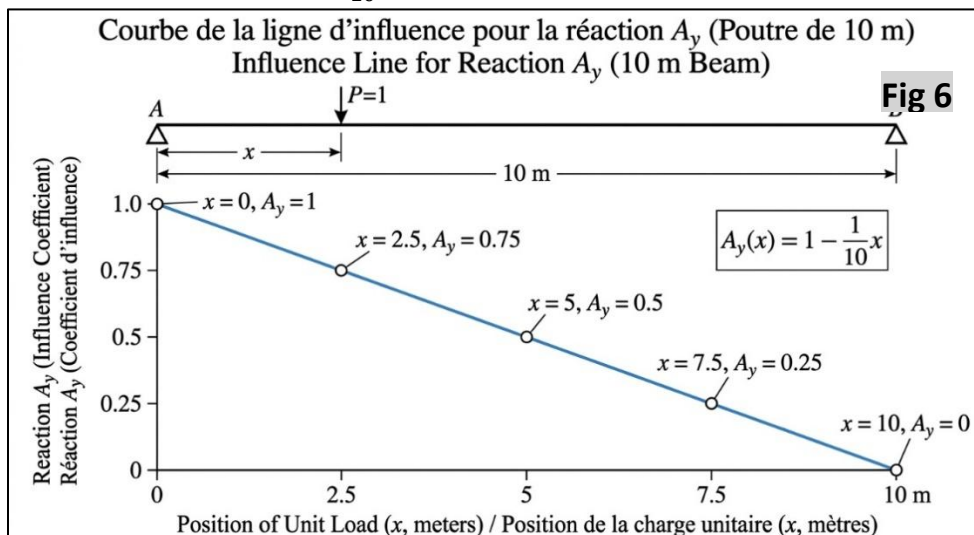
3. **Isolating A_y :**

$$10 \cdot A_y = 10 - x$$

$$A_y = \frac{10 - x}{10} = 1 - \frac{1}{10}x$$

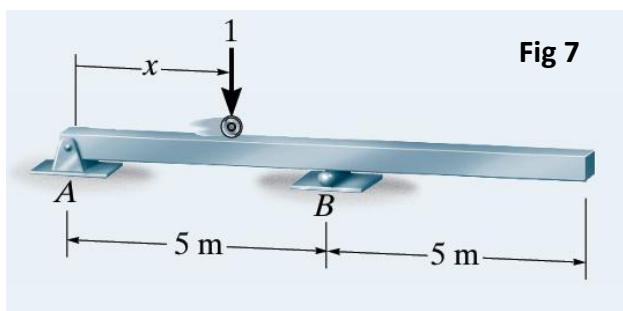
Key Point Verification

- $x = 0 \text{ m}$: $A_y = 1 - \frac{0}{10} = 1$
- $x = 2.5 \text{ m}$: $A_y = 1 - \frac{2.5}{10} = 0.75$
- $x = 5 \text{ m}$: $A_y = 1 - \frac{5}{10} = 0.5$
- $x = 7.5 \text{ m}$: $A_y = 1 - \frac{7.5}{10} = 0.25$
- $x = 10 \text{ m}$: $A_y = 1 - \frac{10}{10} = 0$



Example 02

Construct the influence line for the vertical reaction at B of the beam in **Fig. 7**.



Soltution

Tabulate Values.

Statics verification for the reaction B_y at each load position x using moment equilibrium about support A ($\sum M_A = 0$, counterclockwise positive):

Step-by-Step Statics Calculations

- $x = 0$ m (Support A):

$$\sum M_A = 0 \Rightarrow B_y(5) - 1(0) = 0 \Rightarrow B_y = 0$$

- $x = 2.5$ m (Mid-span AB):

$$\sum M_A = 0 \Rightarrow B_y(5) - 1(2.5) = 0 \Rightarrow B_y = \frac{2.5}{5} = 0.5$$

- $x = 5.0$ m (Support B):

$$\sum M_A = 0 \Rightarrow B_y(5) - 1(5) = 0 \Rightarrow B_y = \frac{5}{5} = 1.0$$

- $x = 7.5$ m (Mid-overhang):

$$\sum M_A = 0 \Rightarrow B_y(5) - 1(7.5) = 0 \Rightarrow B_y = \frac{7.5}{5} = 1.5$$

- $x = 10.0$ m (Free End):

$$\sum M_A = 0 \Rightarrow B_y(5) - 1(10) = 0 \Rightarrow B_y = \frac{10}{5} = 2.0$$

Tabulated Values for B_y

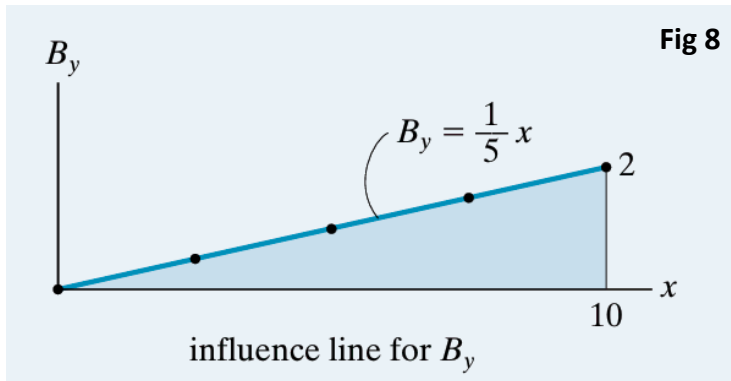
Position x (m)	Moment Equilibrium Equation	Reaction B_y
0.0	$B_y(5) - 1(0) = 0$	0.0
2.5	$B_y(5) - 1(2.5) = 0$	0.5
5.0	$B_y(5) - 1(5) = 0$	1.0
7.5	$B_y(5) - 1(7.5) = 0$	1.5
10.0	$B_y(5) - 1(10) = 0$	2.0

Influence Line Equation for B_y

Influence-Line Equation. We must determine the reaction B_y when the unit load is placed a distance x from A, Fig. 7. Applying moment equilibrium about support A ($\sum M_A = 0$, taking counterclockwise moments as positive) yields the continuous equation for B_y across the entire span Fig 8 : $0 \leq x \leq 10$ m:

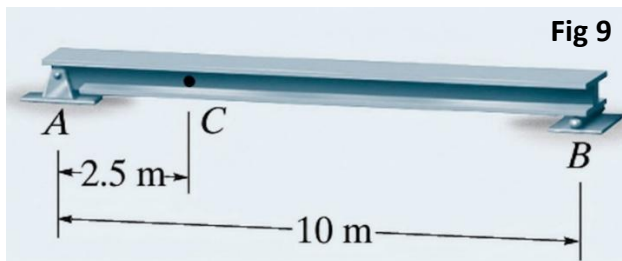
$$\sum M_A = 0 \Rightarrow (B_y \cdot 5) - (1 \cdot x) = 0$$

$$B_y(x) = \frac{x}{5} = 0.2x \quad \text{for } 0 \leq x \leq 10 \text{ m}$$



Example 3

Construct the influence line for the shear at point C of the beam in Fig. 9.



Solution

Tabulate Values. At each selected position x of the unit load, the method of sections is used to calculate the value of V_c . Note in particular that the unit load must be placed just to the left ($x=2.5^-$) and just to the right ($x=2.5^+$) of point C since the shear is discontinuous at C, Figs. 10 and 11. A plot of the values in Table 3 yields the influence line for the shear at C, Fig. 13. Here the diagram gives the shear at C as the unit load moves from one position to the next along the beam.

Internal shear force V_c calculation using static equilibrium equations ($\sum M_A = 0$ and $\sum F_y = 0$) for each unit load position along the beam ($L = 10 \text{ ft} = 3.048 \text{ m}$) with section cut at $x_c = 2.5 \text{ ft} = 0.762 \text{ m}$:

1. Load at $x = 0 \text{ ft} (0 \text{ m})$

- **Support Reactions:**

$$\sum M_A = 0 \Rightarrow B_y(10) - 1(0) = 0 \Rightarrow B_y = 0$$

$$\sum F_y = 0 \Rightarrow A_y + B_y - 1 = 0 \Rightarrow A_y = 1$$

- **Equilibrium of Left Segment AC:**

$$\sum F_y = 0 \Rightarrow A_y - 1 - V_c = 0 \Rightarrow 1 - 1 - V_c = 0 \Rightarrow V_c = 0$$

2. Load at $x = 2.5^- \text{ ft} (0.762^- \text{ m})$ — Just to the left of point C

- **Support Reactions:**

$$\sum M_A = 0 \Rightarrow B_y(10) - 1(2.5) = 0 \Rightarrow B_y = 0.25$$

$$\sum F_y = 0 \Rightarrow A_y = 0.75$$

- **Equilibrium of Right Segment CB:**

$$\sum F_y = 0 \Rightarrow V_C + B_y = 0 \Rightarrow V_C + 0.25 = 0 \Rightarrow V_C = -0.25$$

3. Load at $x = 2.5^+ \text{ ft } (0.762^+ \text{ m})$ — Just to the right of point C

• Support Reactions:

$$\sum M_A = 0 \Rightarrow B_y(10) - 1(2.5) = 0 \Rightarrow B_y = 0.25$$

$$\sum F_y = 0 \Rightarrow A_y = 0.75$$

• Equilibrium of Left Segment AC :

$$\sum F_y = 0 \Rightarrow A_y - V_C = 0 \Rightarrow 0.75 - V_C = 0 \Rightarrow V_C = +0.75$$

4. Load at $x = 5 \text{ ft } (1.524 \text{ m})$

• Support Reactions:

$$\sum M_A = 0 \Rightarrow B_y(10) - 1(5) = 0 \Rightarrow B_y = 0.50$$

$$\sum F_y = 0 \Rightarrow A_y = 0.50$$

• Equilibrium of Left Segment AC :

$$\sum F_y = 0 \Rightarrow A_y - V_C = 0 \Rightarrow 0.50 - V_C = 0 \Rightarrow V_C = +0.50$$

5. Load at $x = 7.5 \text{ ft } (2.286 \text{ m})$

• Support Reactions:

$$\sum M_A = 0 \Rightarrow B_y(10) - 1(7.5) = 0 \Rightarrow B_y = 0.75$$

$$\sum F_y = 0 \Rightarrow A_y = 0.25$$

• Equilibrium of Left Segment AC :

$$\sum F_y = 0 \Rightarrow A_y - V_C = 0 \Rightarrow 0.25 - V_C = 0 \Rightarrow V_C = +0.25$$

6. Load at $x = 10 \text{ ft } (3.048 \text{ m})$

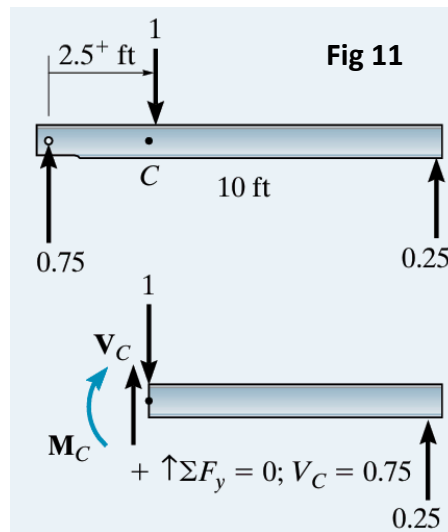
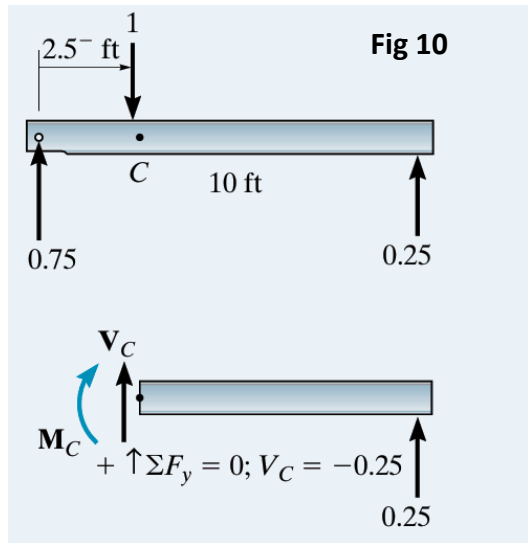
• Support Reactions:

$$\sum M_A = 0 \Rightarrow B_y(10) - 1(10) = 0 \Rightarrow B_y = 1.00$$

$$\sum F_y = 0 \Rightarrow A_y = 0$$

• Equilibrium of Left Segment AC :

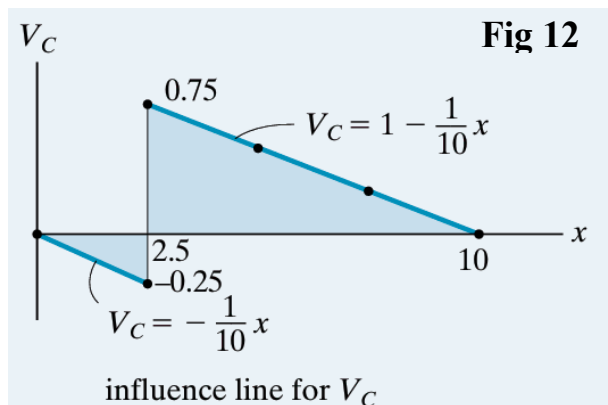
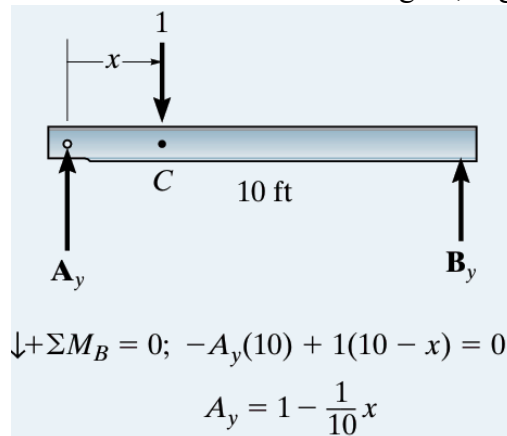
$$\sum F_y = 0 \Rightarrow A_y - V_C = 0 \Rightarrow 0 - V_C = 0 \Rightarrow V_C = 0$$

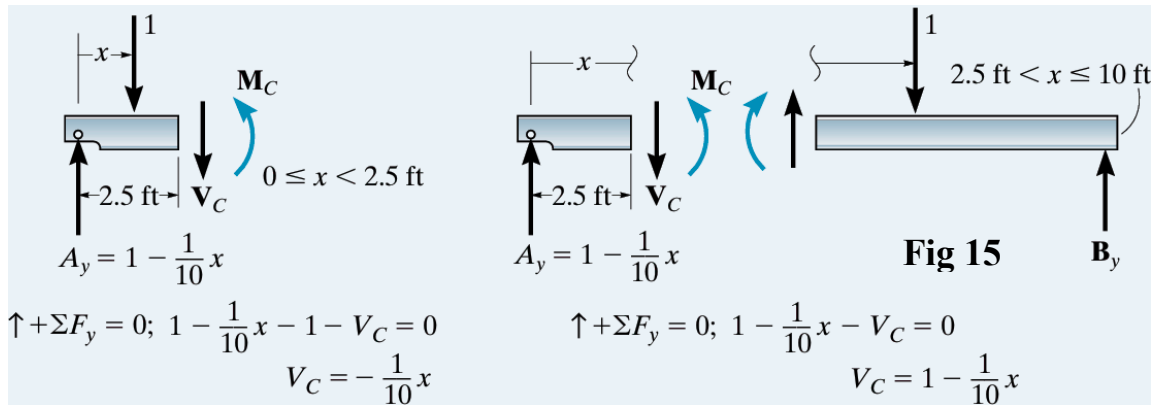


x	V_C
0	0
2.5 ⁻	-0.25
2.5 ⁺	0.75
5	0.5
7.5	0.25
10	0

Influence-Line Equations.

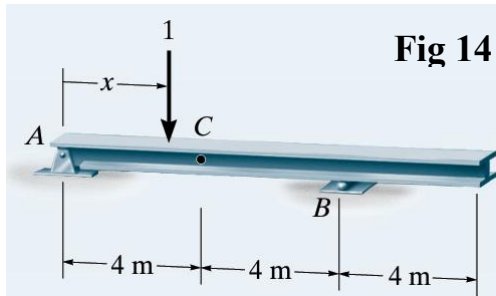
Here two equations have to be determined, since there are two segments for the influence line due to the discontinuity of shear at C. First we determine A_y , Fig 12. Using the result we then obtain the internal shear at C for each region, Fig. 13. The equations are plotted in Fig.14.





Example 04

Construct the influence line for the shear at point C of the beam in Fig. 14



Tabulate values

Calculs Point par Point

- $x = 0$ m (Load at support A / Charge en A) :
 - $\sum M_B = 0 \Rightarrow R_A \cdot 8 - 1 \cdot 8 = 0 \Rightarrow R_A = 1,00$
 - $\sum F_y = 0 \Rightarrow R_B = 1 - 1,00 = 0,00$
 - Shear force / Effort tranchant : $V_C = R_A - 1 = 1,00 - 1 = 0,00$
- $x = 2$ m (Load between A and C / Charge entre A et C) :
 - $\sum M_B = 0 \Rightarrow R_A \cdot 8 - 1 \cdot 6 = 0 \Rightarrow R_A = 0,75$
 - $\sum F_y = 0 \Rightarrow R_B = 1 - 0,75 = 0,25$
 - $V_C = R_A - 1 = 0,75 - 1 = -0,25$
- $x = 4^-$ m (Load just left of C / Charge à gauche de C) :
 - $R_A = 0,50, R_B = 0,50, V_C = R_A - 1 = -0,50$
- $x = 4^+$ m (Load just right of C / Charge à droite de C) :
 - $R_A = 0,50, R_B = 0,50, V_C = R_A = +0,50$
- $x = 6$ m (Load between C and B / Charge entre C et B) :
 - $R_A = 0,25, R_B = 0,75, V_C = R_A = +0,25$
- $x = 8$ m (Load at support B / Charge en B) :
 - $R_A = 0,00, R_B = 1,00, V_C = 0,00$
- $x = 10$ m (Load on overhang / Charge sur le porte-à-faux) :
 - $\sum M_A = 0 \Rightarrow R_B \cdot 8 - 1 \cdot 10 = 0 \Rightarrow R_B = 1,25$
 - $R_A = 1 - 1,25 = -0,25 \Rightarrow V_C = -0,25$
- $x = 12$ m (Load at free end / Charge à l'extrémité libre) :
 - $R_B = 1,50, R_A = -0,50 \Rightarrow V_C = -0,50$

Tableau Récapitulatif (Summary Table)

Position x (m)	Reaction RA	Reaction RB	Shear Force VC
0	1.00	0.00	0.00
2	0.75	0.25	-0.25
4 ⁻	0.50	0.50	-0.50
4 ⁺	0.50	0.50	+0.50
6	0.25	0.75	+0.25
8	0.00	1.00	0.00
10	-0.25	1.25	-0.25
12	-0.50	1.50	-0.50

Influence-Line Equations.

Support Reactions Equations (Équations des réactions d'appui)

- Moment equilibrium about B ($\sum M_B = 0$) [Équilibre des moments en B] :

$$R_A(x) \cdot 8 - 1 \cdot (8 - x) = 0 \Rightarrow R_A(x) = 1 - \frac{x}{8}$$

- Vertical force equilibrium ($\sum F_y = 0$) [Équilibre des forces verticales] :

$$R_A(x) + R_B(x) - 1 = 0 \Rightarrow R_B(x) = \frac{x}{8}$$

Influence Line Equations for Shear Force V_C (Équations de l'effort tranchant en C)

- Segment 1 ($0 \leq x < 4$ m) — Unit load to the left of C (Charge à gauche de C) :

$$V_C(x) = R_A(x) - 1 = \left(1 - \frac{x}{8}\right) - 1 = -\frac{x}{8}$$

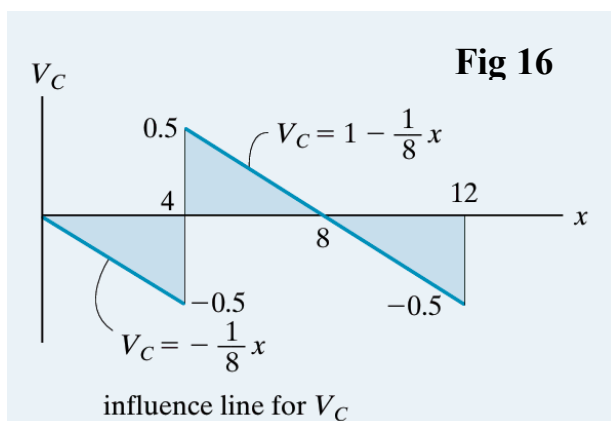
- Segment 2 ($4 < x \leq 12$ m) — Unit load to the right of C (Charge à droite de C) :

$$V_C(x) = R_A(x) = 1 - \frac{x}{8}$$

CHAPTER 01 INFLUENCE LINES OF DETERMINATE AND INDETERMINATE STRUCTURES

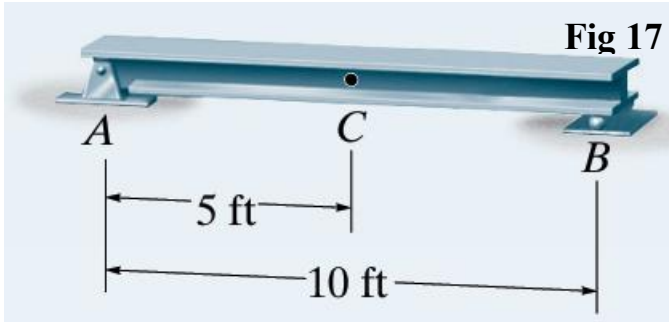
Summary Table (Tableau des valeurs)

Position x (m)	Segment / Condition	Equation for VC	Calculation	Shear Force VC
0	$0 \leq x < 4$	$V_C = -\frac{x}{8}$	$-\frac{0}{8}$	0.00
2	$0 \leq x < 4$	$V_C = -\frac{x}{8}$	$-\frac{2}{8}$	-0.25
4^-	Just left of C	$V_C = -\frac{x}{8}$	$-\frac{4}{8}$	-0.50
4^+	Just right of C	$V_C = 1 - \frac{x}{8}$	$1 - \frac{4}{8}$	+0.50
6	$4 < x \leq 12$	$V_C = 1 - \frac{x}{8}$	$1 - \frac{6}{8}$	+0.25
8	$4 < x \leq 12$	$V_C = 1 - \frac{x}{8}$	$1 - \frac{8}{8}$	0.00
10	$4 < x \leq 12$	$V_C = 1 - \frac{x}{8}$	$1 - \frac{10}{8}$	-0.25
12	$4 < x \leq 12$	$V_C = 1 - \frac{x}{8}$	$1 - \frac{12}{8}$	-0.50



Example 05

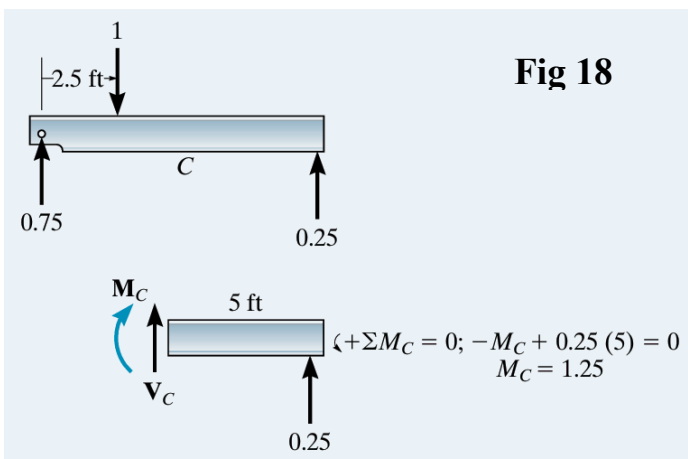
Construct the influence line for the moment at point C of the beam in Fig. 17.



Solution

Tabulate values

At each selected position of the unit load, the value of MC is calculated using the method of sections. For example, see Fig. 18 for x=2.5 ft. A plot of the values in Table 04 yields the influence line for the moment at C figure 19. This diagram gives the moment at C as the unit load moves from one position to the next along the beam.



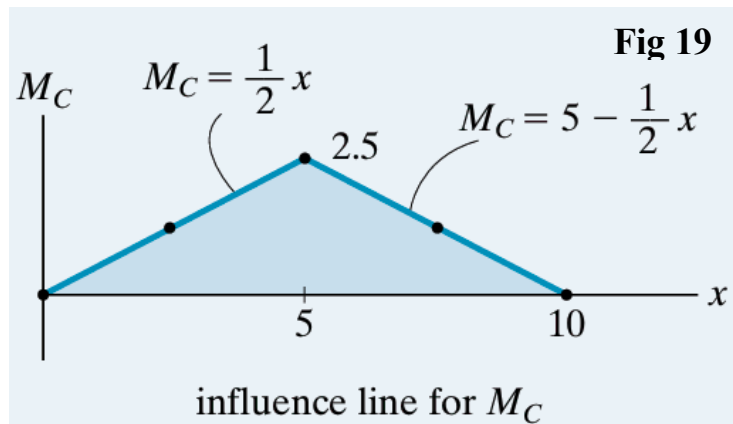
Direct Point-by-Point Calculations (Calculs directs point par point)

- $x = 0$ ft (Load at support A / Charge en A) :
 - $\sum M_B = 0 \Rightarrow R_A \cdot 10 - 1 \cdot 10 = 0 \Rightarrow R_A = 1.00$
 - $\sum F_y = 0 \Rightarrow R_B = 1 - 1.00 = 0.00$
 - Bending moment / Moment fléchissant : $M_C = R_B \cdot 5 = 0.00 \cdot 5 = 0.00$ ft
- $x = 2.5$ ft (Load between A and C / Charge entre A et C) :
 - $\sum M_B = 0 \Rightarrow R_A \cdot 10 - 1 \cdot 7.5 = 0 \Rightarrow R_A = 0.75$
 - $\sum F_y = 0 \Rightarrow R_B = 1 - 0.75 = 0.25$
 - Bending moment / Moment fléchissant : $M_C = R_B \cdot 5 = 0.25 \cdot 5 = 1.25$ ft
- $x = 5$ ft (Load at point C / Charge en C) :
 - $\sum M_B = 0 \Rightarrow R_A \cdot 10 - 1 \cdot 5 = 0 \Rightarrow R_A = 0.50$
 - $\sum F_y = 0 \Rightarrow R_B = 1 - 0.50 = 0.50$
 - Bending moment / Moment fléchissant : $M_C = R_A \cdot 5 = 0.50 \cdot 5 = 2.50$ ft
- $x = 7.5$ ft (Load between C and B / Charge entre C et B) :
 - $\sum M_B = 0 \Rightarrow R_A \cdot 10 - 1 \cdot 2.5 = 0 \Rightarrow R_A = 0.25$

- $\sum F_y = 0 \Rightarrow R_B = 1 - 0.25 = 0.75$
- Bending moment / Moment fléchissant : $M_C = R_A \cdot 5 = 0.25 \cdot 5 = 1.25$ ft
- $x = 10$ ft (Load at support B / Charge en B) :
 - $\sum M_B = 0 \Rightarrow R_A \cdot 10 - 1 \cdot 0 = 0 \Rightarrow R_A = 0.00$
 - $\sum F_y = 0 \Rightarrow R_B = 1 - 0.00 = 1.00$
 - Bending moment / Moment fléchissant : $M_C = R_A \cdot 5 = 0.00 \cdot 5 = 0.00$ ft

Summary Table 04 (Tableau Récapitulatif)

Position x (ft)	Reaction RA	Reaction RB	Bending Moment MC (ft)
0	1.00	0.00	0.00
2.5	0.75	0.25	1.25
5	0.50	0.50	2.50
7.5	0.25	0.75	1.25
10	0.00	1.00	0.00



Influence-Line Equations.

The two line segments for the influence line can be determined using $\sum M_C = 0$ along with the method of sections shown in Fig. 19.

Support Reactions Equations

- Moment equilibrium about support B ($\sum M_B = 0$) :

$$R_A(x) \cdot 10 - 1 \cdot (10 - x) = 0 \Rightarrow R_A(x) = 1 - \frac{x}{10}$$
- Vertical force equilibrium ($\sum F_y = 0$) :

$$R_A(x) + R_B(x) - 1 = 0 \Rightarrow R_B(x) = \frac{x}{10}$$

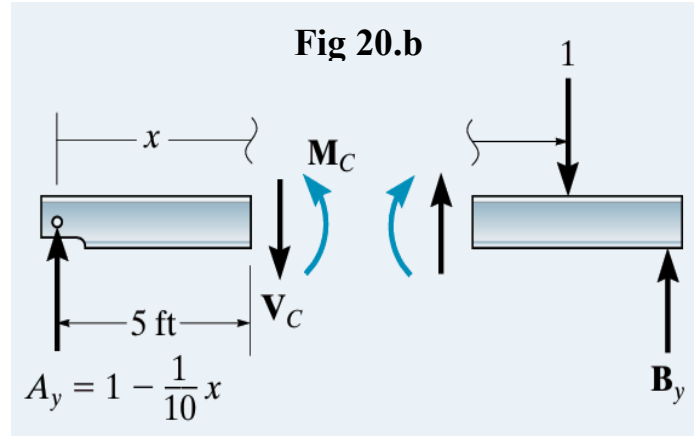
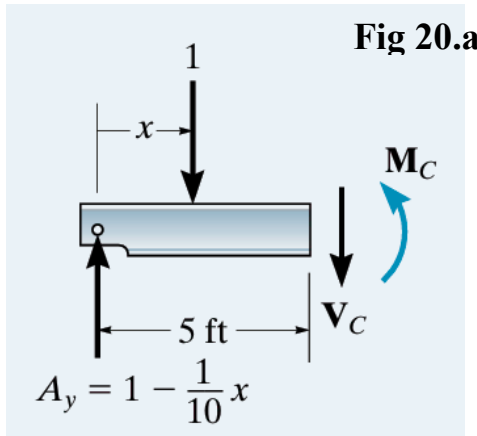
Bending Moment Equations for M_C (Équations du moment fléchissant en C)

- Segment 1 ($0 \leq x \leq 5$ ft) — Unit load to the left of C (figure 20.a):

$$M_C(x) = R_B(x) \cdot 5 = \left(\frac{x}{10}\right) \cdot 5 = 0.5x$$

- Segment 2 ($5 \leq x \leq 10$ ft) — Unit load to the right of C (figure 20. b) :

$$M_C(x) = R_A(x) \cdot 5 = \left(1 - \frac{x}{10}\right) \cdot 5 = 5 - 0.5x$$

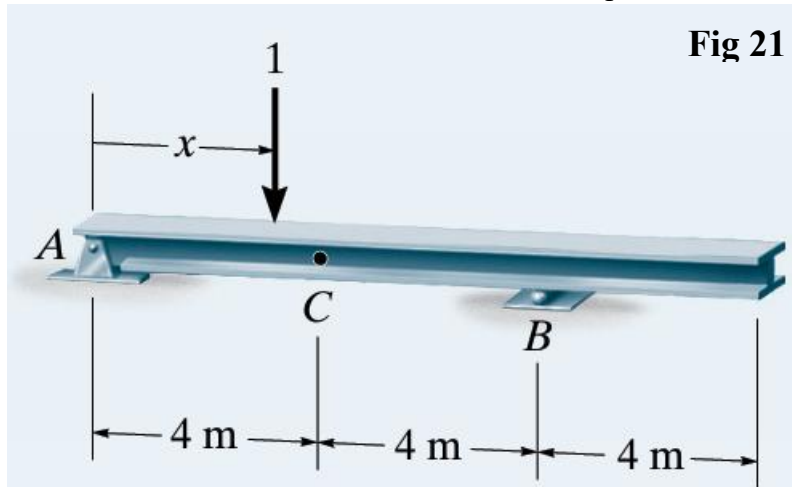


Summary Table (Tableau récapitulatif par équations)

Position x (ft)	Segment / Condition	Equation for MC	Calculation	Reaction RA	Reaction RB	Bending Moment MC (ft)
0	$0 \leq x \leq 5$	$M_C = 0.5x$	$0.5 \cdot 0$	1.00	0.00	0.00
2.5	$0 \leq x \leq 5$	$M_C = 0.5x$	$0.5 \cdot 2.5$	0.75	0.25	1.25
5	Point C	$M_C = 0.5x$	$0.5 \cdot 5$	0.50	0.50	2.50
7.5	$5 \leq x \leq 10$	$M_C = 5 - 0.5x$	$5 - 0.5 \cdot 7.5$	0.25	0.75	1.25
10	$5 \leq x \leq 10$	$M_C = 5 - 0.5x$	$5 - 0.5 \cdot 10$	0.00	1.00	0.00

Example 06

Construct the influence line for the moment at point C of the beam in Fig. 21



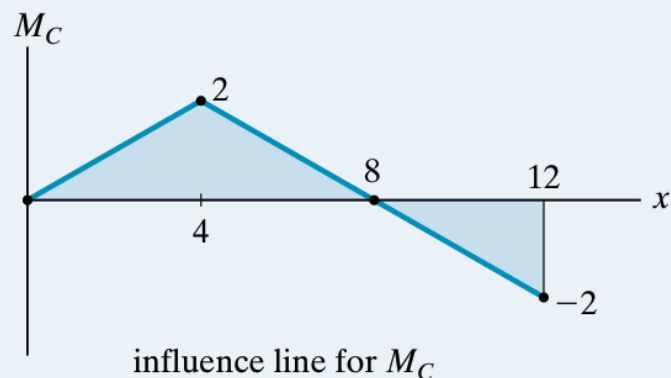
SOLUTION

Tabulate Values. Using statics and the method of sections, verify that the values of the moment M_C at point C in Fig. 21 correspond to each position x of the unit load. A plot of the values in Fig. 22.b yields the influence line in Fig. 22.c.

Fig 22

x	M_C
0	0
4	2
8	0
12	-2

(b)



(c)

Influence-Line Equations.

Support Reactions Equations (Équations des réactions d'appui)

- **Moment equilibrium about B ($\sum M_B = 0$) [Équilibre des moments en B] :**

$$R_A(x) \cdot 8 - 1 \cdot (8 - x) = 0 \Rightarrow R_A(x) = 1 - \frac{x}{8}$$

- **Vertical force equilibrium ($\sum F_y = 0$) [Équilibre des forces verticales] :**

$$R_A(x) + R_B(x) - 1 = 0 \Rightarrow R_B(x) = \frac{x}{8}$$

Influence Line Equations for Bending Moment M_C (Équations du moment fléchissant)

- **Segment 1 ($0 \leq x \leq 4$ m) — Unit load to the left of C (Charge à gauche de C) :**

$$M_C(x) = R_B(x) \cdot 4 = \left(\frac{x}{8}\right) \cdot 4 = 0.5x$$

- **Segment 2 ($4 \leq x \leq 12$ m) — Unit load to the right of C (Charge à droite de C) :**

$$M_C(x) = R_A(x) \cdot 4 = \left(1 - \frac{x}{8}\right) \cdot 4 = 4 - 0.5x$$

Summary Table (Tableau Récapitulatif)

Position x (m)	Segment Condition /	Equation for MC	Reaction RA	Reaction RB	Bending Moment MC (m)
0	$0 \leq x \leq 4$	$M_C = 0.5 \cdot 0$	1.00	0.00	0.00
2	$0 \leq x \leq 4$	$M_C = 0.5 \cdot 2$	0.75	0.25	1.00
4	Point C	$M_C = 0.5 \cdot 4$	0.50	0.50	2.00
6	$4 \leq x \leq 12$	$M_C = 4 - 0.5 \cdot 6$	0.25	0.75	1.00
8	Support B	$M_C = 4 - 0.5 \cdot 8$	0.00	1.00	0.00
10	Overhang (Console)	$M_C = 4 - 0.5 \cdot 10$	-0.25	1.25	-1.00
12	Free End (Extrémité libre)	$M_C = 4 - 0.5 \cdot 12$	-0.50	1.50	-2.00

I.3 MÜLLER-BRESLAU PRINCIPLE : QUALITATIVE KINEMATIC APPROACH

In 1886, Heinrich Müller-Breslau developed a technique for quickly determining the **shape of an influence line**.

The Müller-Breslau principle is based on the **principle of virtual work** and provides a simple way to construct the qualitative shape of influence lines.

Principle

The qualitative shape of the influence line for a given function, such as:

- **Support reaction (R)**

- Shear force (T) or (V)
- Bending moment (M)

can be obtained by:

1. **Releasing the restraint** corresponding to the function being studied at the section where the influence line is required.
2. **Applying a unit action** corresponding to this function in its **positive direction**.
3. Allowing the modified structure to **deform**.

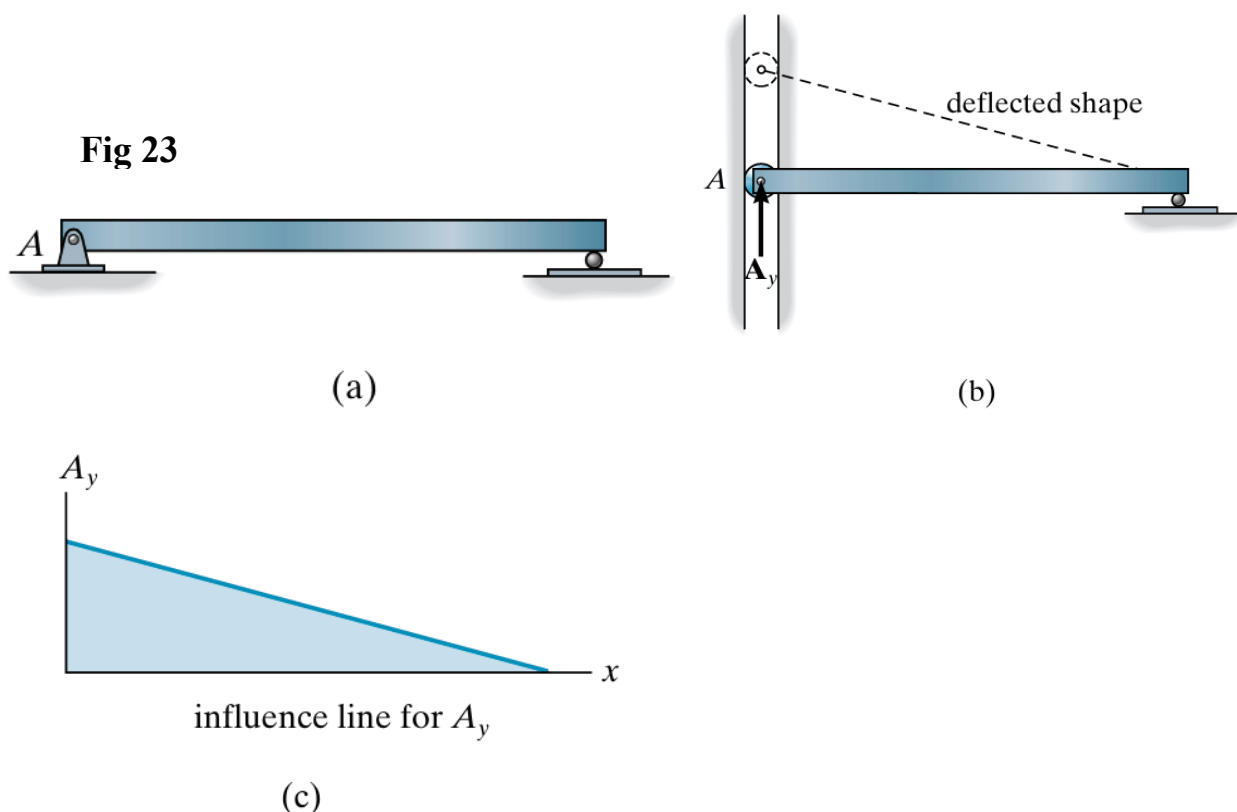
The deflected shape of the modified structure represents the qualitative shape of the influence line.

Easy rule to remember

RELEASE → APPLY → DEFORM → INFLUENCE LINE

Example 1: Influence Line for the Vertical Reaction (A_y)

Consider the beam shown in Fig. 23.a.



We want to determine the **qualitative shape of the influence line for the vertical reaction (A_y) at A**.

Step 1 – Release the reaction

Replace the **pin** at A with a **roller guide**, as shown in Fig. 23.b.

The roller guide still allows the beam to resist the **horizontal force**, but it cannot resist the **vertical reaction (A_y)**.

Step 2 – Apply the positive action

Apply a **positive upward force (A_y)** at A.

Step 3 – Observe the deformation

The beam deflects into the **dashed position** shown in Fig. 23.c.

Conclusion

The dashed deflected shape in Fig. 23.c represents the qualitative shape of the influence line for (A_y).

Remember:

Release (A_y) → Apply (+ A_y) → Observe the deflected shape → Influence line for (A_y).

Example 2: Influence Line for the Shear Force at C

Let us determine the qualitative shape of the influence line for the shear force (V_C) at section (C), as shown in Fig. 24.a.

Step 1 – Release the shear

The section at (C) is replaced by a **roller guide**, as shown in Fig. 24.b.

This support can resist **moment and axial force**, but it cannot resist **shear force**.

Step 2 – Apply the positive shear

Apply a **positive shear force (V_C)** at point (C).

Step 3 – Observe the deformation

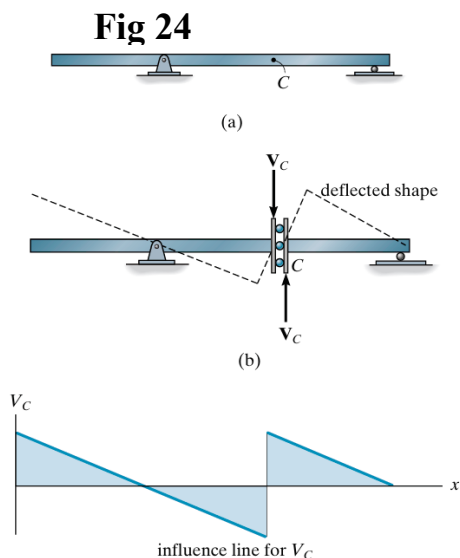
The beam deforms into the **dotted shape** shown in Fig. 24.b.

Step 4 – Obtain the influence line

This **deformed shape** gives the qualitative shape of the **influence line for (V_C)**, as shown in Fig. 24.c.

Conclusion

Release the shear → Apply (+ V_C) → Observe the deformation → Obtain the influence line for (V_C).



Example 3: Influence Line for the Bending Moment at C

Let us determine the qualitative shape of the influence line for the bending moment M_C at section C, as shown in Fig. 25.a.

Step 1 – Release the moment

Introduce an **internal hinge (articulation)** at C, as shown in Fig. 25.b.

The internal hinge can resist **axial and shear forces**, but it **cannot resist a bending moment**.

Step 2 – Apply the positive moment

Apply a **positive bending moment M_C** at C.

Step 3 – Observe the deformation

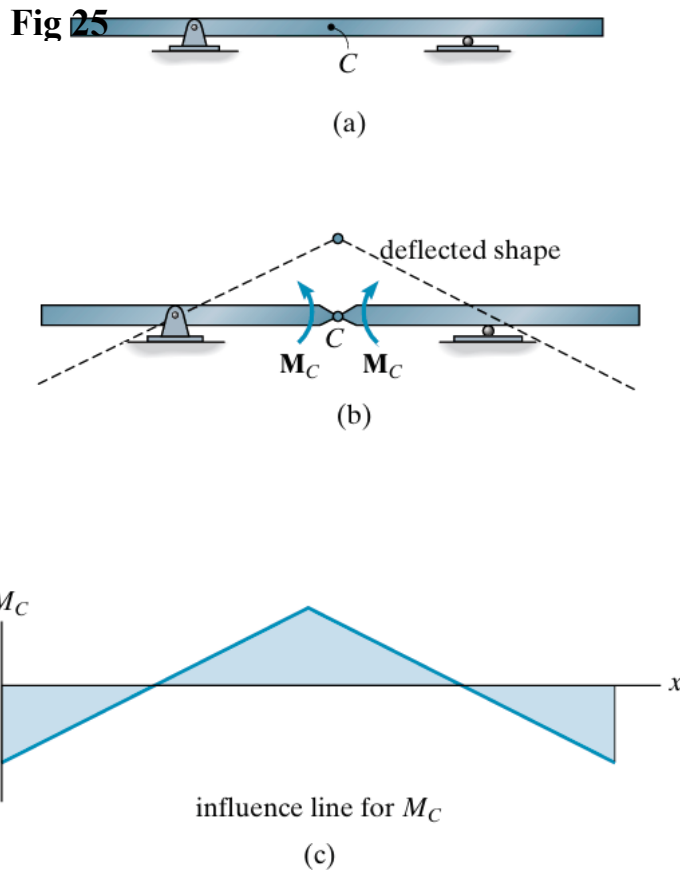
The beam deforms into the **dotted position** shown in Fig. 25.b.

Step 4 – Obtain the influence line

The resulting **deformed shape** represents the qualitative shape of the **influence line for M_C** , as shown in Fig. 25.c.

Conclusion

Release the moment $\rightarrow$ Apply $+M_C \rightarrow$ Observe the deformation $\rightarrow$ Obtain the influence line for M_C .



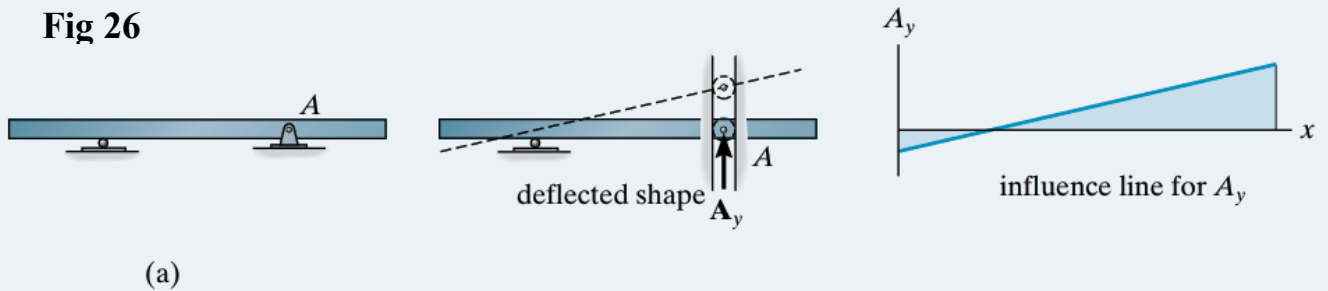
Exmaple 4

For each beam in Fig. 26.a through 26.c, sketch the influence line for the vertical reaction at A

SOLUTION

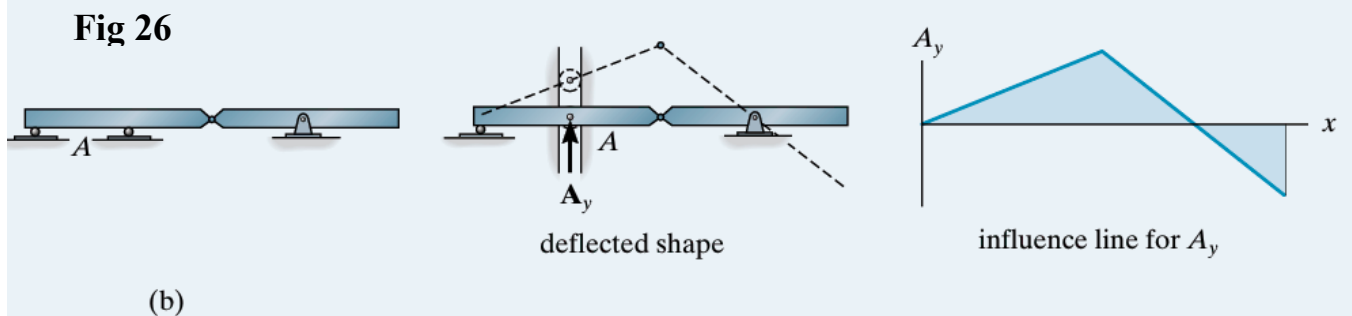
The support is replaced by a roller guide at A since it will resist A_x , but not A_y . The force A_y is then applied.

Fig 26



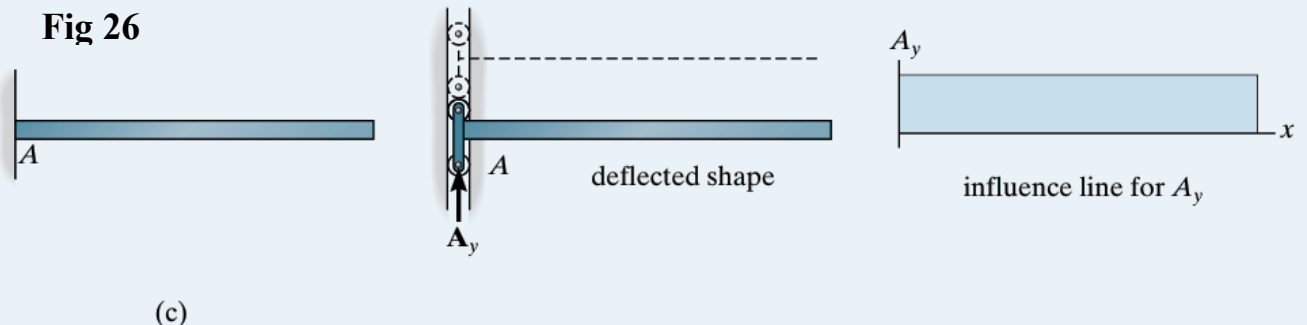
Again, a roller guide is placed at A and the force A_y is applied

Fig 26



A double-roller guide must be used at A in this case, since this type of support will resist both a moment M_A at the fixed support and axial load A_x , but will not resist A_y .

Fig 26



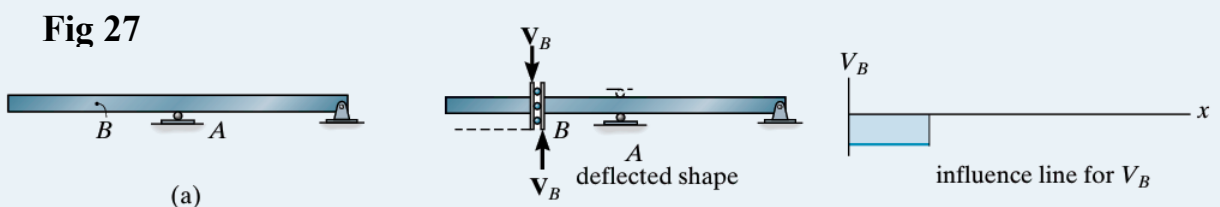
Example 05

For each beam in Figs. 27.a through 27.c, sketch the influence line for the shear at B.

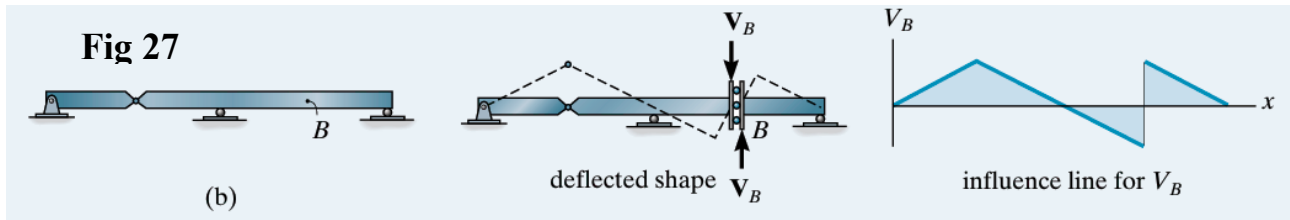
SOLUTION

The roller guide is introduced at B and the positive shear V_B is applied. Notice that the right segment of the beam will not deflect since the roller at A actually constrains the beam from moving vertically, either up or down

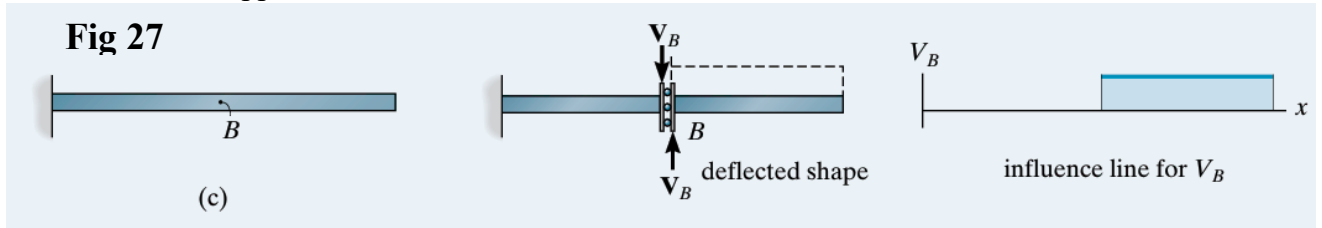
Fig 27



Placing the roller guide at B and applying the positive shear at B yields the deflected shape and corresponding influence line.



Again, the roller guide is placed at B, the positive shear is applied, and the deflected shape and corresponding influence line are shown. Note that the left segment of the beam does not deflect, due to the fixed support.

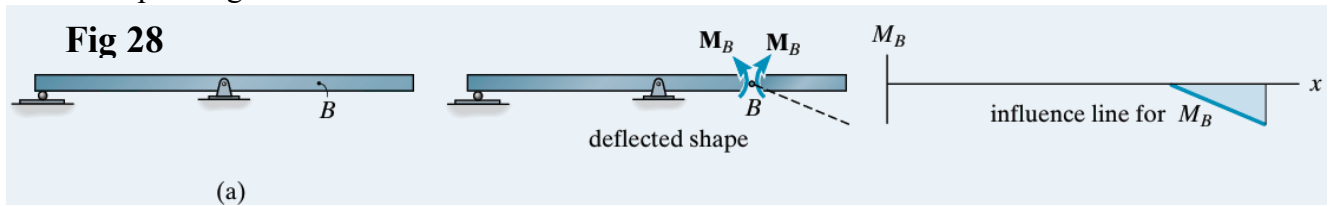


Example 06

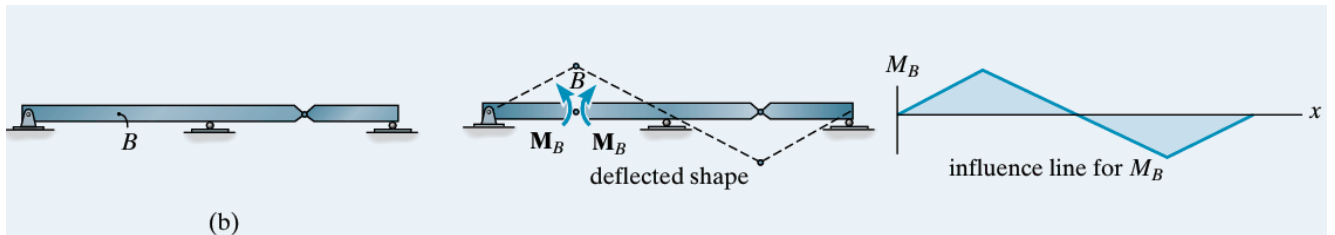
For each beam in Figs. 28.a through Fig. 28.c, sketch the influence line for the moment at B.

SOLUTION

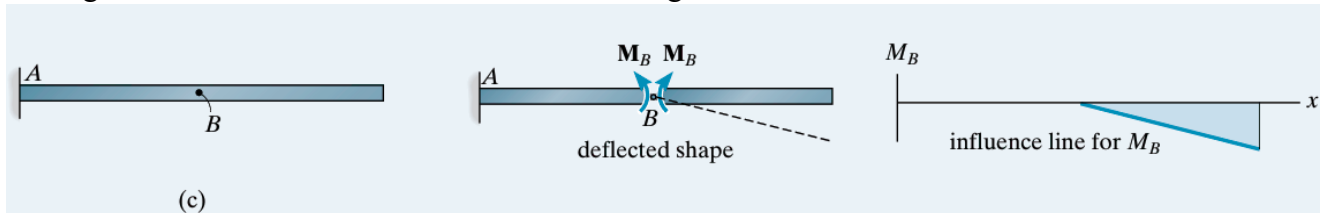
A hinge is introduced at B and positive moments M_B are applied to the beam. The deflected shape and corresponding influence line are shown.



Placing a hinge at B and applying positive moments M_B to the beam yields the deflected shape and influence line.

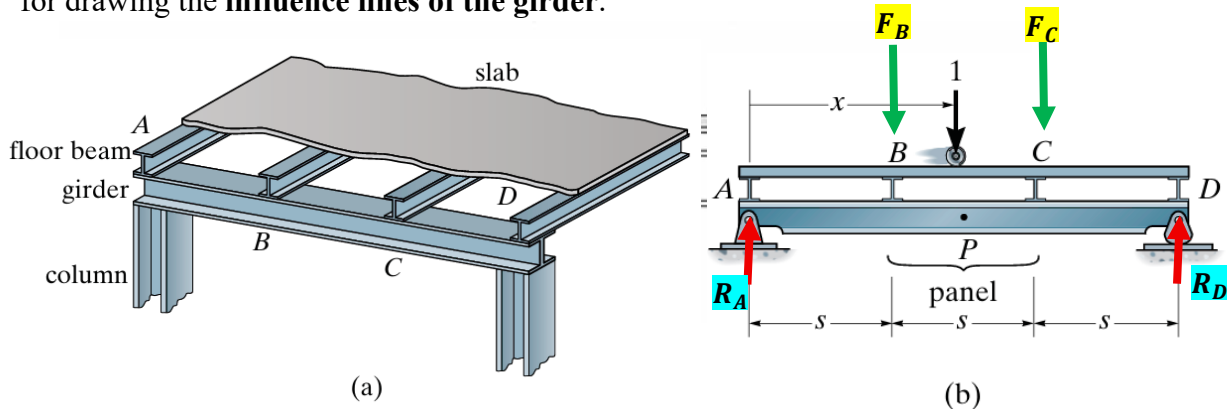


With the hinge and positive moment at B the deflected shape and influence line are shown. The left segment of the beam is constrained from moving due to the fixed wall at A



1.4. Influence Lines for Floor Girders

In a floor system, loads are transferred from the **slab** to the **floor beams**, then to the **girder**, and finally to the **columns**. The girder is simply supported by the columns. As shown in **Fig. 29.a**, the load is transferred to the girder at specific points **A, B, C, and D**, called **panel points**. The part between two panel points, such as **B–C** in **Fig. 29.b**, is called a **panel**. These points are important for drawing the **influence lines of the girder**.



The influence line for a specified point on the girder can be determined using the same statics procedure, place the unit load at various points x on the floor slab and always compute the function (shear or moment) at the specified point P in the girder, **Fig. 29.b**. Plotting these values versus x yields the influence line for the function at P .

For example, when the unit load is placed on the floor slab, as shown in **Fig. 29.b**:

1. Determine the slab reactions F_B and F_C .
2. Use these reactions to calculate the girder support reactions F_A and F_D .
3. Cut the girder at point P and use the **method of sections** to determine the internal shear V_P and internal moment M_P .

$$V_P = R_A - F_B$$

$$M_P = R_A x - F_B(x-s)$$

where:

- V_P = internal shear at point P
- M_P = internal moment at point P
- R_A = girder reaction at A
- F_B = reaction from the floor slab at B
- x = distance from A to P
- s = distance from A to B

Important : V_P is constant within the panel BC , whereas M_P depends on the exact position of P .

Step-by-Step Analysis

1. Slab Reactions (F_B and F_C)

For a unit load acting on slab panel BC at distance x from A :

- Span of the panel segment = s
- Position of unit load relative to point $B = x - s$

Applying static equilibrium to the slab span BC :

$$\sum M_C = 0 \Rightarrow F_B \cdot s = 1 \cdot (2s - x)$$

$$F_B = \frac{2s - x}{s} = 2 - \frac{x}{s}$$

$$\sum M_B = 0 \Rightarrow F_C \cdot s = 1 \cdot (x - s)$$

$$F_C = \frac{x - s}{s} = \frac{x}{s} - 1$$

Girder Support Reactions (R_A and R_D)

The total length of the girder is $L = 3s$. The loads transmitted from the slab through the floor beams onto the girder are F_B at point B ($x = s$) and F_C at point C ($x = 2s$).

Applying equilibrium to the entire girder:

- Taking moments about D ($x = 3s$):

$$\sum M_D = 0 \Rightarrow R_A(3s) = F_B(2s) + F_C(s)$$

$$R_A(3s) = \left(2 - \frac{x}{s}\right)(2s) + \left(\frac{x}{s} - 1\right)(s) = 3s - x$$

$$R_A = \frac{3s - x}{3s} = 1 - \frac{x}{3s}$$

- Taking moments about A ($x = 0$):

$$\sum M_A = 0 \Rightarrow R_D(3s) = F_B(s) + F_C(2s)$$

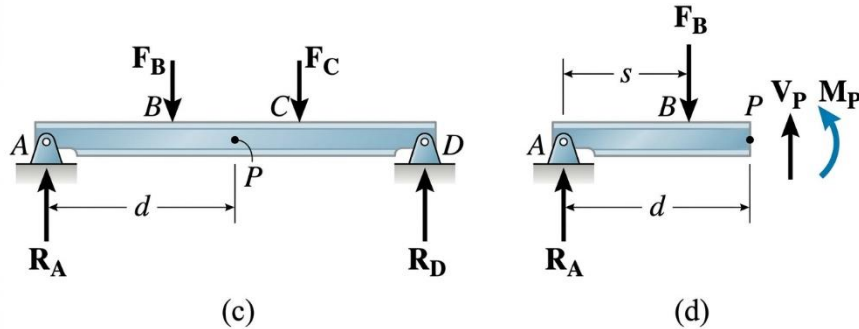
$$R_D(3s) = \left(2 - \frac{x}{s}\right)(s) + \left(\frac{x}{s} - 1\right)(2s) = x$$

$$R_D = \frac{x}{3s}$$

4. Internal Forces at Point P (V_P and M_P)

CHAPTER 01 INFLUENCE LINES OF DETERMINATE AND INDETERMINATE STRUCTURES

Consider a section cut at point P located at distance d from support A (where $s < d < 2s$). Isolate the left segment of the girder (A to P) Fig29.c and Fig 29.d:



- **Internal Shear (V_P):**

$$\sum F_y = 0 \Rightarrow R_A - F_B - V_P = 0$$

$$V_P = R_A - F_B = \left(1 - \frac{x}{3s}\right) - \left(2 - \frac{x}{s}\right) = -1 + \frac{2x}{3s}$$

(Note: V_P depends only on the unit load position x , remaining uniform for any point P within panel BC .)

- **Internal Moment (M_P):**

$$\sum M_P = 0 \Rightarrow M_P = R_A \cdot d - F_B \cdot (d - s)$$

$$M_P = \left(1 - \frac{x}{3s}\right)d - \left(2 - \frac{x}{s}\right)(d - s)$$

$$M_P = 2s - d - x + \frac{2xd}{3s}$$

Here is the calculation summary for the three unit load positions (measuring distance x from support A):

Case 1: Unit Load at Point B ($x = s$)

- **Slab Reactions:**

$$\circ F_B = 2 - \frac{s}{s} = 1$$

$$\circ F_C = \frac{s}{s} - 1 = 0$$

- **Girder Support Reactions:**

$$\circ R_A = 1 - \frac{s}{3s} = \frac{2}{3}$$

$$\circ R_D = \frac{s}{3s} = \frac{1}{3}$$

- **Internal Forces at Point P :**

- Shear force: $V_P = -1 + \frac{2(s)}{3s} = -\frac{1}{3}$
- Bending moment: $M_P = \left(\frac{2}{3}\right)d - 1(d - s) = s - \frac{1}{3}d$

Intermediate Case: Unit Load Between B and C (Midpoint of Panel, $x = 1.5s = \frac{3}{2}s$)

The midpoint of panel BC is located at distance $x = s + 0.5s = 1.5s$ from support A.

• **Slab Reactions:**

- $F_B = 2 - \frac{1.5s}{s} = \frac{1}{2}$
- $F_C = \frac{1.5s}{s} - 1 = \frac{1}{2}$

• **Girder Support Reactions:**

- $R_A = 1 - \frac{1.5s}{3s} = \frac{1}{2}$
- $R_D = \frac{1.5s}{3s} = \frac{1}{2}$

• **Internal Forces at Point P:**

- Shear force: $V_P = R_A - F_B = \frac{1}{2} - \frac{1}{2} = 0$
- Bending moment: $M_P = \left(\frac{1}{2}\right)d - \frac{1}{2}(d - s) = \frac{1}{2}s$

Case 2: Unit Load at Point C ($x = 2s$)

• **Slab Reactions:**

- $F_B = 2 - \frac{2s}{s} = 0$
- $F_C = \frac{2s}{s} - 1 = 1$

• **Girder Support Reactions:**

- $R_A = 1 - \frac{2s}{3s} = \frac{1}{3}$
- $R_D = \frac{2s}{3s} = \frac{2}{3}$

• **Internal Forces at Point P:**

- Shear force: $V_P = -1 + \frac{2(2s)}{3s} = \frac{1}{3}$
- Bending moment: $M_P = \left(\frac{1}{3}\right)d - 0(d - s) = \frac{1}{3}d$

What this means for the Influence Line (IL) Because the transmitted panel loads F_B and F_C vary linearly (as straight lines) between B and C, any structural effect in the main girder (support

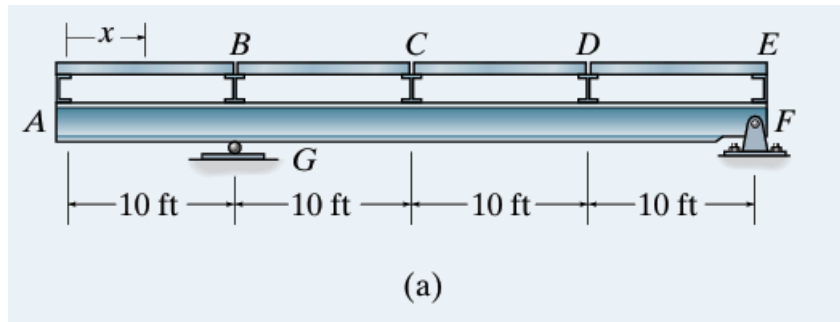
reactions, shear force V_p , or bending moment M_p) must also vary linearly as a straight line across panel BC

💡 The Golden Rule in Practice

- Thanks to this fundamental property, you never need to set up equations for load positions in the middle of a panel! To draw a complete Influence Line for an indirectly loaded floor system: Calculate the effect only when the unit load is placed at each panel point (A, B, C, D, etc.). Plot these specific values on your diagram. Connect adjacent points using straight line segments.
- As long as the load remains in panel CD , panel BC receives no direct transferred load. The internal shear passing through this entire panel remains completely constant from B to C , equal to the support reaction R_A .

Example 01

Draw the influence line for the shear in panel CD of the floor girder in **Fig. 30.a**.



Derivation of Support Reactions R_G and R_F

The main girder rests on a roller support at G ($x = 10$ ft) and a pin support at F ($x = 40$ ft). The distance between supports is $L_{GF} = 40 - 10 = 30$ ft.

To derive the support reaction formulas for any unit load position x measured from point A :

- Sum of moments about support F ($x = 40$ ft):**

$$\sum M_F = 0 \Rightarrow R_G \cdot (40 - 10) - 1 \cdot (40 - x) = 0$$

$$30R_G = 40 - x \Rightarrow R_G = \frac{40 - x}{30}$$

- Sum of moments about support G ($x = 10$ ft):**

$$\sum M_G = 0 \Rightarrow R_F \cdot (40 - 10) - 1 \cdot (x - 10) = 0$$

$$30R_F = x - 10 \Rightarrow R_F = \frac{x - 10}{30}$$

- Vertical Equilibrium Check:**

$$R_G + R_F = \frac{40 - x}{30} + \frac{x - 10}{30} = \frac{30}{30} = 1 \quad (\text{Satisfied})$$

2. General Expression for Panel Shear V_{CD}

To find the internal shear force V_{CD} , make a section cut anywhere inside panel CD ($20 \text{ ft} < x_{\text{cut}} < 30 \text{ ft}$) and sum the vertical forces acting to the left of the cut:

$$\sum F_y = 0 \Rightarrow R_G - F_A - F_B - F_C - V_{CD} = 0$$

$$V_{CD} = R_G - (F_A + F_B + F_C)$$

3. Detailed Step-by-Step Calculations

Case 1: Load at Node A ($x = 0 \text{ ft}$)

- Reactions:

$$R_G = \frac{40 - 0}{30} = \frac{40}{30} = \frac{4}{3}$$

$$R_F = \frac{0 - 10}{30} = -\frac{10}{30} = -\frac{1}{3}$$

- Transmitted Floor Loads: $F_A = 1, F_B = 0, F_C = 0$
- Panel Shear:

$$V_{CD} = R_G - F_A = \frac{4}{3} - 1 = +\frac{1}{3}$$

Case 2: Load at Node B ($x = 10 \text{ ft}$)

- Reactions:

$$R_G = \frac{40 - 10}{30} = \frac{30}{30} = 1$$

$$R_F = \frac{10 - 10}{30} = 0$$

- Transmitted Floor Loads: $F_A = 0, F_B = 1, F_C = 0$
- Panel Shear:

$$V_{CD} = R_G - F_B = 1 - 1 = 0$$

Case 3: Load at Node C ($x = 20 \text{ ft}$)

- Reactions:

$$R_G = \frac{40 - 20}{30} = \frac{20}{30} = \frac{2}{3}$$

$$R_F = \frac{20 - 10}{30} = \frac{10}{30} = \frac{1}{3}$$

- **Transmitted Floor Loads:** $F_A = 0, F_B = 0, F_C = 1$
- **Panel Shear:**

$$V_{CD} = R_G - F_C = \frac{2}{3} - 1 = -\frac{1}{3}$$

Case 4: Load at Midpoint of Panel CD ($x = 25$ ft)

- **Reactions:**

$$R_G = \frac{40 - 25}{30} = \frac{15}{30} = \frac{1}{2}$$

$$R_F = \frac{25 - 10}{30} = \frac{15}{30} = \frac{1}{2}$$

- **Transmitted Floor Loads (Distributed across Panel CD):**

$$F_C = \frac{30 - 25}{10} = \frac{5}{10} = \frac{1}{2}$$

$$F_D = \frac{25 - 20}{10} = \frac{5}{10} = \frac{1}{2}$$

$$F_A = 0, \quad F_B = 0$$

- **Panel Shear:**

$$V_{CD} = R_G - F_C = \frac{1}{2} - \frac{1}{2} = 0$$

Case 5: Load at Node D ($x = 30$ ft)

- **Reactions:**

$$R_G = \frac{40 - 30}{30} = \frac{10}{30} = \frac{1}{3}$$

$$R_F = \frac{30 - 10}{30} = \frac{20}{30} = \frac{2}{3}$$

- **Transmitted Floor Loads:**

$$F_A = 0, \quad F_B = 0, \quad F_C = 0, \quad F_D = 1$$

- **Panel Shear:**

(Since F_D acts to the right of the section cut, it does not enter the left-hand section equilibrium)

$$V_{CD} = R_G - 0 = +\frac{1}{3}$$

Case 6: Load at Node E ($x = 40$ ft)

- **Reactions:**

$$R_G = \frac{40 - 40}{30} = 0$$

$$R_F = \frac{40 - 10}{30} = \frac{30}{30} = 1$$

- **Transmitted Floor Loads:**

$$F_A = 0, \quad F_B = 0, \quad F_C = 0, \quad F_E = 1$$

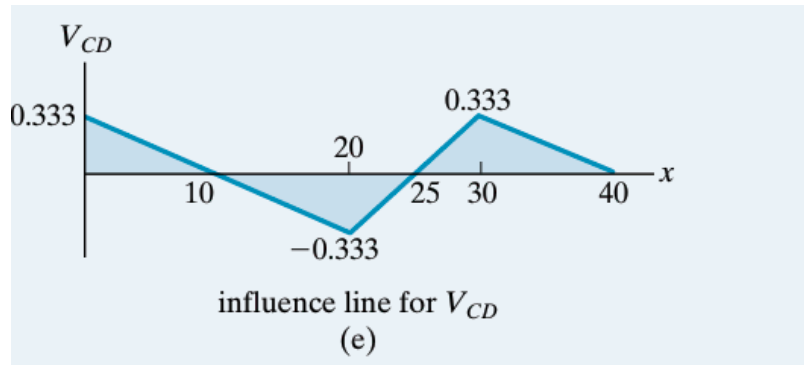
- **Panel Shear:**

$$V_{CD} = R_G - 0 = 0$$

4. Summary Table

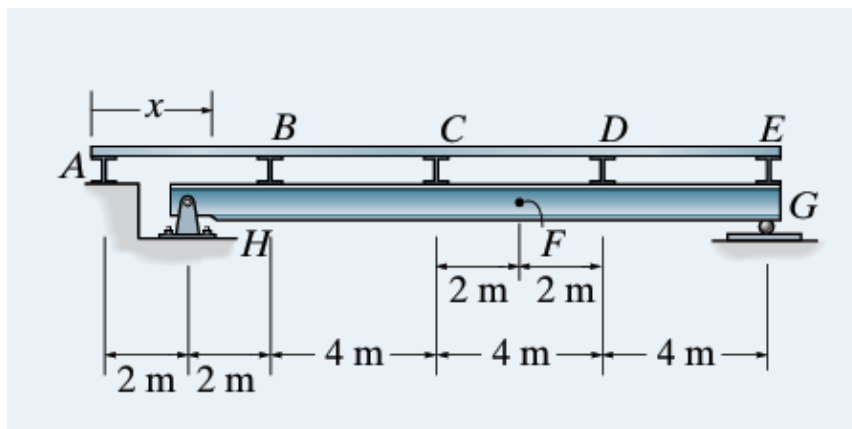
Load Position x	Panel Point	R_G	R_F	F_C	Panel Shear V_{CD}
0 ft	A	$+4/3$	$-1/3$	0	$+1/3$
10 ft	B (G)	1	0	0	0
20 ft	C	$+2/3$	$+1/3$	1	$-1/3$
25 ft	Midpoint CD	$+1/2$	$+1/2$	$\frac{1}{2}$	0
30 ft	D	$+1/3$	$+2/3$	0	$+1/3$
40 ft	E (F)	0	1	0	0

Influence Line. When the tabular values are plotted and the points connected with straight line segments, the resulting influence line for V_{CD} is as shown in Fig. 30.e.



Example 2

Draw the influence line for the moment at point F for the floor girder in Fig. 31.



Solution

Static Model and System Geometry

From the structural diagram:

- **Origin ($x = 0$):** Set at the leftmost end at node A.
- **Panel Points (Floor Beams / Load Transfer Nodes):**
 - Node A: $x_A = 0$ m (rests directly on the abutment/wall)
 - Node B: $x_B = 4$ m
 - Node C: $x_C = 8$ m
 - Node D: $x_D = 12$ m
 - Node E: $x_E = 16$ m
- **Main Girder Supports:**
 - Pin support H: $x_H = 2$ m
 - Roller support G: $x_G = 16$ m

○ Main girder span: $L = x_G - x_H = 16 - 2 = 14$ m

- **Section of Interest F :** $x_F = 10$ m (located 8 m from H and 6 m from G).

2. General Support Reaction Equations

For a concentrated force P acting on the main girder at position x_{girder} (measured from A):

- **Sum of moments about support H ($x = 2$ m):**

$$\sum M_H = 0 \Rightarrow R_G \cdot (14) - P \cdot (x_{\text{girder}} - 2) = 0 \Rightarrow R_G = P \cdot \frac{x_{\text{girder}} - 2}{14}$$

- **Sum of moments about support G ($x = 16$ m):**

$$\sum M_G = 0 \Rightarrow R_H \cdot (14) - P \cdot (16 - x_{\text{girder}}) = 0 \Rightarrow R_H = P \cdot \frac{16 - x_{\text{girder}}}{14}$$

3. Principle of Indirect Loading

In an indirectly loaded floor system, the unit moving load (1 kN) travels across the slab, which transfers concentrated loads exclusively to the **panel points** (A, B, C, D, E).

Between adjacent panel points, any internal structural effect varies **strictly linearly** (straight line segments). Therefore, we only need to calculate M_F by placing the 1 kN unit load successively at each panel point.

4. Node-by-Node Detailed Calculations

Case 1: Unit Load at Node A ($x = 0$ m)

- The load is applied directly above the abutment wall.
- No force is transmitted to the main girder ($F_B = F_C = F_D = F_E = 0$).
- Support reactions: $R_H = 0, R_G = 0$.
- **Moment at F :**

$$M_F = 0 \text{ kN} \cdot \text{m}$$

Case 2: Unit Load at Node B ($x = 4$ m)

- The floor slab transfers the entire load to node B : $F_B = 1$ kN at $x = 4$ m.
- Support reaction at G :

$$R_G = 1 \cdot \frac{4 - 2}{14} = \frac{2}{14} = \frac{1}{7} \text{ kN}$$

- Cut section at point F ($x_F = 10$ m) and consider the right segment ($F \rightarrow G$):

$$M_F = R_G \cdot (16 - 10) = \left(\frac{1}{7}\right) \cdot 6 = +\frac{6}{7} \text{ kN} \cdot \text{m} \approx +0.857 \text{ kN} \cdot \text{m}$$

Case 3: Unit Load at Node C ($x = 8$ m)

- The floor slab transfers the entire load to node C : $F_C = 1$ kN at $x = 8$ m.
- Support reaction at G :

$$R_G = 1 \cdot \frac{8 - 2}{14} = \frac{6}{14} = \frac{3}{7} \text{ kN}$$

- Cut section at point F ($x_F = 10$ m) and consider the right segment ($F \rightarrow G$):

$$M_F = R_G \cdot (16 - 10) = \left(\frac{3}{7}\right) \cdot 6 = +\frac{18}{7} \text{ kN} \cdot \text{m} \approx +2.571 \text{ kN} \cdot \text{m}$$

Case 4: Unit Load at Node D ($x = 12$ m)

- The floor slab transfers the entire load to node D : $F_D = 1$ kN at $x = 12$ m.
- Support reaction at H :

$$R_H = 1 \cdot \frac{16 - 12}{14} = \frac{4}{14} = \frac{2}{7} \text{ kN}$$

- Cut section at point F ($x_F = 10$ m) and consider the left segment ($H \rightarrow F$):

$$M_F = R_H \cdot (10 - 2) = \left(\frac{2}{7}\right) \cdot 8 = +\frac{16}{7} \text{ kN} \cdot \text{m} \approx +2.286 \text{ kN} \cdot \text{m}$$

Case 5: Unit Load at Node E ($x = 16$ m)

- The load acts directly over support G ($R_G = 1$ kN, $R_H = 0$).
- Cut section at point F and consider the left segment:

$$M_F = R_H \cdot (10 - 2) = 0 \cdot 8 = 0 \text{ kN} \cdot \text{m}$$

5. Verification at Intermediate Position ($x = 10$ m)

When the unit load is directly over point F ($x = 10$ m, midpoint of panel CD):

- Load distribution to nodes C and D :

$$F_C = \frac{12 - 10}{4} = \frac{1}{2} \text{ kN}, \quad F_D = \frac{10 - 8}{4} = \frac{1}{2} \text{ kN}$$

- Support reaction at G :

$$R_G = \left(\frac{1}{2}\right) \cdot \frac{8 - 2}{14} + \left(\frac{1}{2}\right) \cdot \frac{12 - 2}{14} = \frac{3}{14} + \frac{5}{14} = \frac{8}{14} = \frac{4}{7} \text{ kN}$$

- Moment at F (right segment $F \rightarrow G$):

$$M_F = R_G \cdot (16 - 10) - F_D \cdot (12 - 10) = \left(\frac{4}{7}\right) \cdot 6 - \left(\frac{1}{2}\right) \cdot 2 = \frac{24}{7} - 1 = +\frac{17}{7} \text{ kN} \cdot \text{m} \\ \approx +2.429 \text{ kN} \cdot \text{m}$$

(Note: This matches the arithmetic mean between $M_F(C) = \frac{18}{7}$ and $M_F(D) = \frac{16}{7}$, confirming linear continuity across the panel).

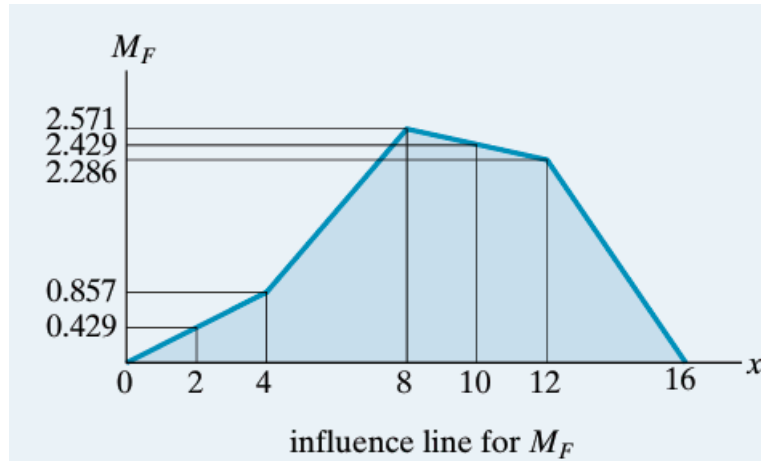
6. Summary Table

Load Position (x)	Panel Point	Reaction RH	Reaction RG	Moment MF (kN·m)
$x = 0$ m	A	0	0	0
$x = 4$ m	B	$+6/7$	$+1/7$	$+\frac{6}{7} \approx +0.857$
$x = 8$ m	C	$+4/7$	$+3/7$	$+\frac{18}{7} \approx +2.571$ (Maximum)
$x = 10$ m	Midpoint CD (F)	$+3/7$	$+4/7$	$+\frac{17}{7} \approx +2.429$
$x = 12$ m	D	$+2/7$	$+5/7$	$+\frac{16}{7} \approx +2.286$
$x = 16$ m	E	0	1	0

7. Plotting Instructions

- Horizontal Baseline:** Draw a reference axis from $x = 0$ to $x = 16$ m marking nodes A, B, C, D, E.
- Plot Key Ordinates:**
 - At A ($x = 0$): 0
 - At B ($x = 4$ m): $+6/7$
 - At C ($x = 8$ m): $+18/7$
 - At D ($x = 12$ m): $+16/7$
 - At E ($x = 16$ m): 0
- Connect Segments:** Join adjacent points sequentially ($A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$) using straight line segments:
 - $A \rightarrow B$: Increasing line from 0 to $+6/7$.
 - $B \rightarrow C$: Increasing line from $+6/7$ to the maximum peak at $+18/7$.

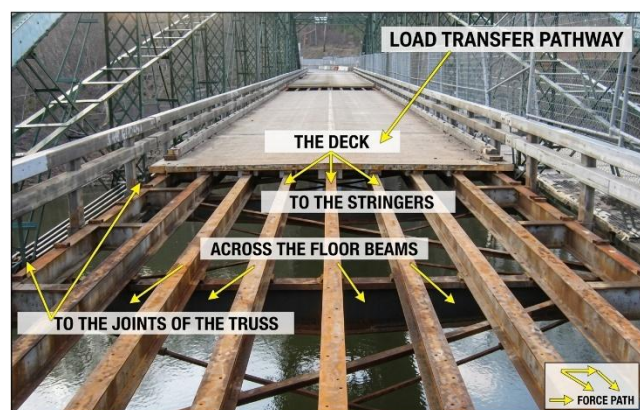
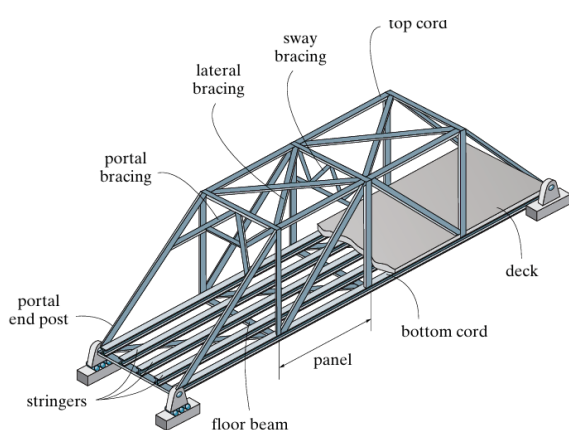
- $C \rightarrow D$: Continuous decreasing line from $+18/7$ to $+16/7$ (passing through $+17/7$ at $x = 10$ m).
- $D \rightarrow E$: Decreasing line from $+16/7$ back to 0.



1.5 Influence Lines for Trusses

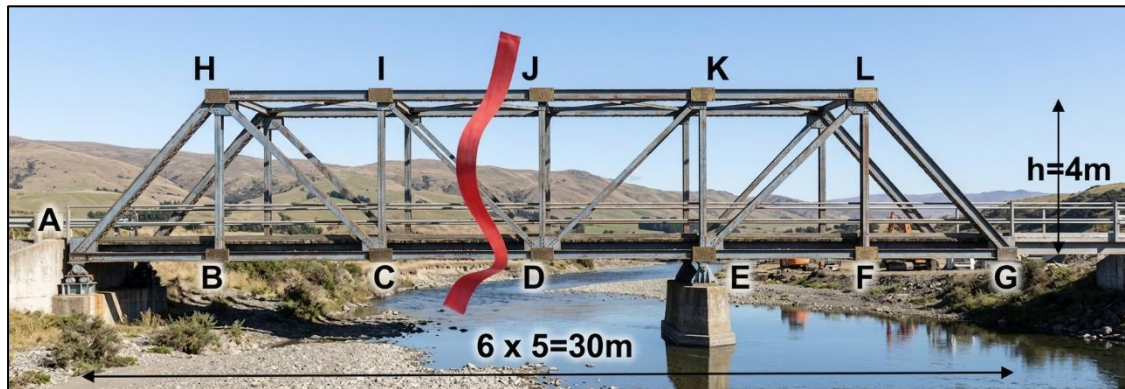
Trusses are often used as primary load-carrying elements for bridges. Hence, for design it is important to be able to construct the influence lines for each of its members. As shown in Fig. 33, the loading on the bridge deck is transmitted to stringers, which in turn transmit the loading to floor beams and then to the joints along the bottom cord of the truss.

Trusses are commonly used to support bridge loads. To draw the **influence line of a truss member**, a **unit load** is placed successively at each joint along the bottom chord. The force in the member is then calculated using the **method of joints** or **method of sections**. The calculated forces are plotted against the load positions. **Tension is taken as positive (+)** and **compression as negative (-)**. Finally, the points are connected with straight lines to obtain the influence line.



Exercise 01

Draw the **influence line diagrams** for the axial forces in members **IJ**, **CD**, and **ID** of the bridge truss shown in Fig. 6-24a.



To plot the influence lines for the axial forces in members *IJ*, *CD*, and *ID*, we use Method of Sections (Ritter's method).

1. Geometry and Support Reactions

- **Dimensions:**
 - 6 panels of 5 m each $\Rightarrow$ Total length = 30 m.
 - Truss height $h = 4$ m.
 - Node coordinates along the bottom chord:
 - $A = 0$ m (Support A)
 - $B = 5$ m
 - $C = 10$ m
 - $D = 15$ m
 - $E = 20$ m (Support E)
 - $F = 25$ m
 - $G = 30$ m (Cantilever end from E to G)
- **Support Reactions for a moving downward unit load $P = 1$ at position $x \in [0, 30]$ m:**

$$\sum M_E = 0 \Rightarrow R_A(x) = \frac{20 - x}{20}$$

$$\sum M_A = 0 \Rightarrow R_E(x) = \frac{x}{20}$$

2. Influence Line for Axial Force N_{IJ} (Top Chord)

Pass a section cutting through members IJ , ID , and CD . Taking moments about node D ($x_D = 15 \text{ m}$, $y_D = 0$):

- **Load to the right of D** ($15 \text{ m} \leq x \leq 30 \text{ m}$):

Isolating the left section:

$$\sum M_D = 0 \Rightarrow R_A \cdot 15 + N_{IJ} \cdot 4 = 0 \Rightarrow N_{IJ}(x) = -\frac{15}{4} R_A(x) = \frac{3(x - 20)}{16}$$

- **Load to the left of D** ($0 \leq x \leq 15 \text{ m}$):

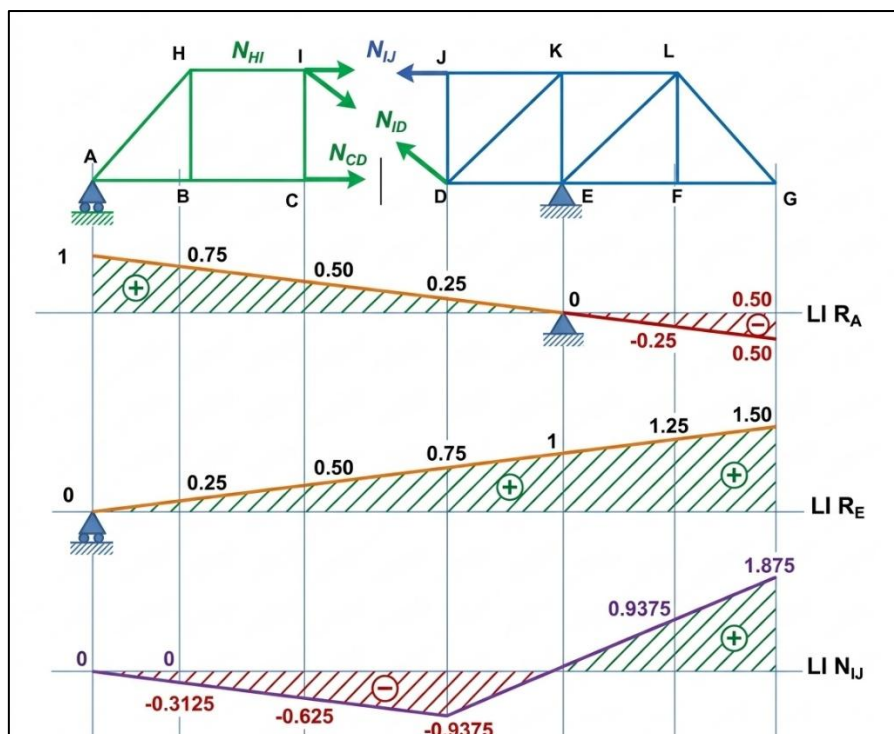
Isolating the right section:

$$\sum M_D = 0 \Rightarrow R_E \cdot 5 + N_{IJ} \cdot 4 = 0 \Rightarrow N_{IJ}(x) = -\frac{5}{4} R_E(x) = -\frac{x}{16}$$

Key Values for N_{IJ} :

Position x (m)	0 (A)	5 (B)	10 (C)	15 (D)	20 (E)	25 (F)	30 (G)
N_{IJ}	0	-0.3125	-0.625	-0.9375	0	+0.9375	+1.875

- **Diagram Shape:** A negative triangular shape from $A(0)$ to $E(20)$ peaking at $D(15)$ with -0.9375 , followed by a straight upward line reaching $+1.875$ at $G(30)$.



3. Influence Line for Axial Force N_{CD} (Bottom Chord)

Taking moments about node I ($x_I = 10$ m, $y_I = 4$ m):

- **Load to the left of C** ($0 \leq x \leq 10$ m):

Isolating the right section:

$$\sum M_I = 0 \Rightarrow R_E \cdot 10 - N_{CD} \cdot 4 = 0 \Rightarrow N_{CD}(x) = \frac{10}{4} R_E(x) = \frac{x}{8}$$

- **Load to the right of C** ($10 \text{ m} \leq x \leq 30 \text{ m}$):

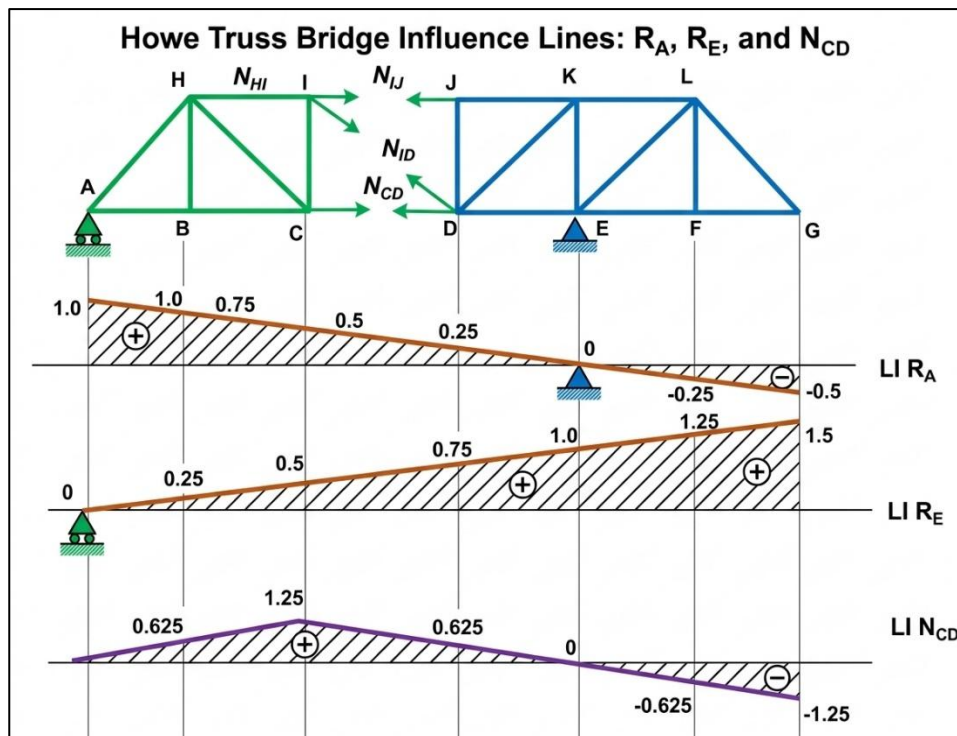
Isolating the left section:

$$\sum M_I = 0 \Rightarrow R_A \cdot 10 - N_{CD} \cdot 4 = 0 \Rightarrow N_{CD}(x) = \frac{10}{4} R_A(x) = \frac{20 - x}{8}$$

Key Values for N_{CD} :

Position x (m)	0 (A)	5 (B)	10 (C)	15 (D)	20 (E)	25 (F)	30 (G)
N_{CD}	0	+0.625	+1.25	+0.625	0	-0.625	-1.25

- **Diagram Shape:** A positive triangle from $A(0)$ to $E(20)$ peaking at $C(10)$ with +1.25, then sloping downward to -1.25 at $G(30)$.



4. Influence Line for Axial Force N_{ID} (Diagonal)

Diagonal ID connects node $I(10,4)$ to $D(15,0)$.

- Inclination: $\sin \alpha = \frac{h}{L_{ID}} = \frac{4}{\sqrt{5^2+4^2}} = \frac{4}{\sqrt{41}} \approx 0.6247$.

Applying vertical force equilibrium $\sum F_y = 0$ across the section:

- Load to the left of panel CD ($0 \leq x \leq 10$ m):**

Isolating the right section:

$$\sum F_y = 0 \Rightarrow R_E + N_{ID} \sin \alpha = 0 \Rightarrow N_{ID}(x) = -\frac{R_E(x)}{\sin \alpha} = -\frac{\sqrt{41}}{80} x \approx -0.08004 x$$

- Load to the right of panel CD ($15 \text{ m} \leq x \leq 30 \text{ m}$):**

Isolating the left section:

$$\sum F_y = 0 \Rightarrow R_A - N_{ID} \sin \alpha = 0 \Rightarrow N_{ID}(x) = \frac{R_A(x)}{\sin \alpha} = \frac{\sqrt{41}}{80} (20 - x) \approx 0.08004 (20 - x)$$

- Inside panel CD ($10 \text{ m} \leq x \leq 15 \text{ m}$):**

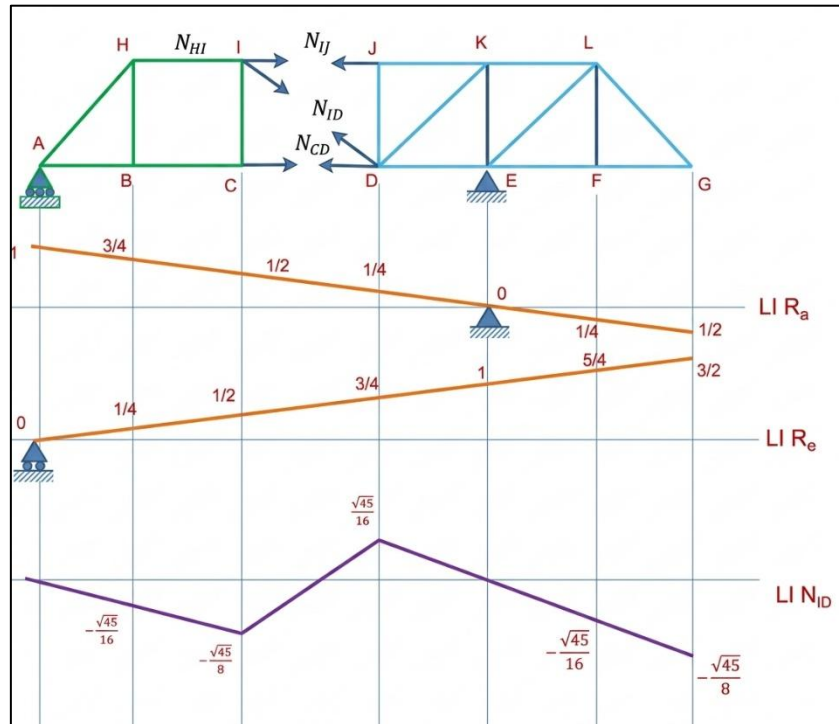
Linear transition between $x = 10 \text{ m}$ and $x = 15 \text{ m}$, crossing zero ($N_{ID} = 0$) at $x_0 = 13.33 \text{ m}$ (i.e., $x = \frac{40}{3} \text{ m}$).

Key Values for N_{ID} :

Position x (m)	0 (A)	10 (C)	13.33	15 (D)	20 (E)	30 (G)
N_{ID} (Exact)	0	$-\frac{\sqrt{41}}{8}$	0	$+\frac{\sqrt{41}}{16}$	0	$-\frac{\sqrt{41}}{8}$
N_{ID} (Decimal)	0	-0.800	0	+0.400	0	-0.800

- Diagram Shape:**

- From $A(0)$ to $C(10)$: linearly decreases from 0 to -0.800 .
- From $C(10)$ to $D(15)$: linear transition line passing through 0 at $x = 13.33 \text{ m}$ up to $+0.400$ at $D(15)$.
- From $D(15)$ to $G(30)$: straight line going through 0 at $E(20)$ and down to -0.800 at $G(30)$.

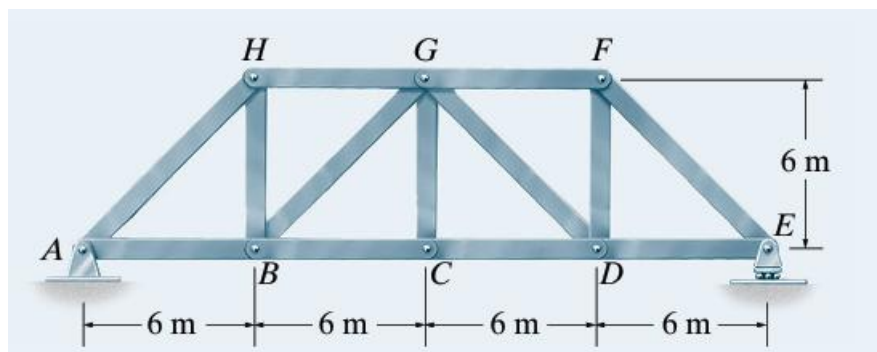


Plotting Summary

Node / Position x	A (0 m)	C (10 m)	D (15 m)	E (20 m)	G (30 m)
N_{IJ}	0	-0.625	-0.938	0	+1.875
N_{CD}	0	+1.250	+0.625	0	-1.250
N_{ID}	0	-0.800	+0.400	0	-0.800

Example 02

Draw the influence line for the force in member CG of the bridge truss shown in Fig. 34.



Solution

. Static Analysis of Joint C (Method of Joints)

To determine the force F_{CG} in the vertical member CG , isolate joint C :

- **Members meeting at joint C:**
 - Two collinear horizontal bottom-chord members: CB and CD .
 - One vertical member: CG .
- **External loading:**
 - As the unit moving load $P = 1$ moves across the bridge deck, a fraction P_C is transferred to joint C via the floor beams.

Summing forces vertically ($\sum F_y = 0$) at joint C :

$$\sum F_y = 0 \Rightarrow F_{CG} - P_C = 0 \Rightarrow F_{CG} = P_C$$

Thus, the axial force in member CG is directly equal to the fraction of the unit load transmitted to joint C .

2. Calculation of Force F_{CG} for Various Load Positions (x)

Consider a unit load $P = 1$ moving along the bottom chord from $x = 0$ m (joint A) to $x = 24$ m (joint E):

1. **Load between $x = 0$ m and $x = 6$ m (Panel AB):**
 - The load is carried entirely by joints A and B .
 - No load is transmitted to joint C ($P_C = 0$).
 - $F_{CG} = 0$
2. **Load between $x = 6$ m and $x = 12$ m (Panel BC):**
 - By the lever rule, the fraction transmitted to joint C increases linearly:

$$P_C = \frac{x - 6}{12 - 6} = \frac{x - 6}{6}$$

- At $x = 6$ m (Joint B): $F_{CG} = 0$
 - At $x = 12$ m (Joint C): $F_{CG} = +1$ (Tension)
3. **Load between $x = 12$ m and $x = 18$ m (Panel CD):**
 - The fraction transmitted to joint C decreases linearly:

$$P_C = \frac{18 - x}{18 - 12} = \frac{18 - x}{6}$$

- At $x = 12$ m (Joint C): $F_{CG} = +1$ (Tension)
- At $x = 18$ m (Joint D): $F_{CG} = 0$

4. Load between $x = 18$ m and $x = 24$ m (Panel DE):

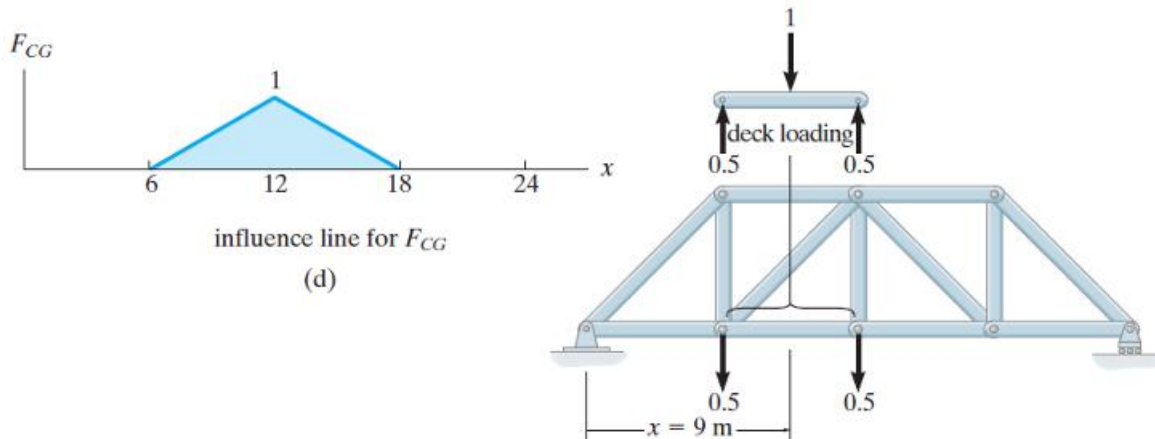
- The load is carried entirely by joints D and E .
- No load is transmitted to joint C ($P_C = 0$).
- $F_{CG} = 0$

3. Summary Table of Ordinates

Load Position x (m)	Node	Force F_{CG}	Stress State
0	A	0	Zero force
6	B	0	Zero force
12	C	+1	Maximum Tension
18	D	0	Zero force
24	E	0	Zero force

4. Instructions for Plotting the Influence Line

- Horizontal Baseline:** Draw a horizontal axis representing the total span from $x = 0$ m to $x = 24$ m, marking nodes A, B, C, D, E .
- Plot Segments:**
 - From $x = 0$ m to $x = 6$ m ($A \rightarrow B$): Draw a horizontal line lying on the axis ($F_{CG} = 0$).
 - From $x = 6$ m to $x = 12$ m ($B \rightarrow C$): Draw a line sloping upward from 0 to a maximum value of +1 above the axis.
 - From $x = 12$ m to $x = 18$ m ($C \rightarrow D$): Draw a line sloping downward from +1 to 0.
 - From $x = 18$ m to $x = 24$ m ($D \rightarrow E$): Draw a horizontal line lying on the axis ($F_{CG} = 0$).



I.6 Influence Lines for frame

For a structural **frame**, influence lines can be used to determine the **support reactions, shear force, and bending moment** at a specific point. A **unit load** ($W = 1$) is moved along the frame, and the corresponding value of the required function is calculated for each load position. These values are then plotted against the load position to obtain the **influence line**.

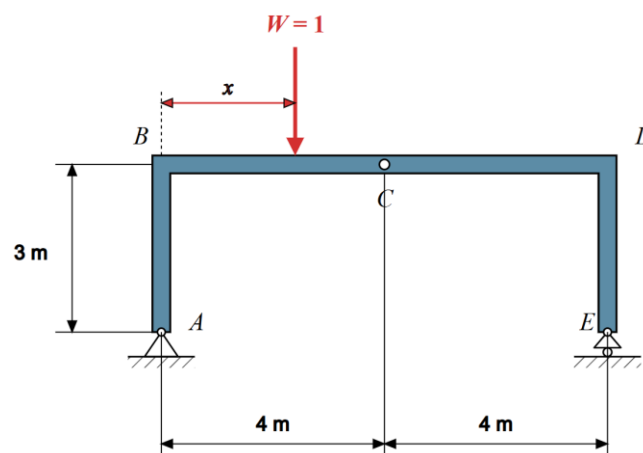
Key Points:

- Move a unit load ($W = 1$) along the frame.
- Calculate the desired function (reaction, shear force, or bending moment) at the specified point.
- Plot the calculated values versus the load position (x).
- Connect the key points with straight lines.

Example 01

Consider the frame shown in **Fig. 36**. The goal is to draw the influence lines for:

- The vertical reaction at support A (R_A)
- The shear force at point C (V_C)
- The bending moment at point C (M_C)



Solution

Global Static Analysis & Reaction Equations

- Frame geometry and supports :
 - The frame consists of a horizontal beam BD with a total span of $L = 8$ m (4 m + 4 m) and two vertical columns of height $h = 3$ m.
 - Support A is a pin support (restraining vertical reaction R_A and horizontal reaction H_A).
 - Support E is a roller support (restraining vertical reaction R_E only, $H_E = 0$).
 - A downward unit vertical load $W = 1$ moves along the horizontal member BD , with its position defined by distance x measured from joint B ($0 \leq x \leq 8$ m).
- Static equilibrium equations:
 - Horizontal Equilibrium ($\sum F_x = 0$):

$$H_A = 0$$

Since no horizontal external force acts on the frame, the columns carry no horizontal shear, and the frame behaves like a simply supported beam of 8 m span.

- Moment Equilibrium about Support E ($\sum M_E = 0$):

$$R_A \cdot 8 - 1 \cdot (8 - x) = 0 \Rightarrow R_A(x) = 1 - \frac{x}{8}$$

- Vertical Equilibrium ($\sum F_y = 0$):

$$R_A + R_E = 1 \Rightarrow R_E(x) = \frac{x}{8}$$

2. Influence Line Equations & Calculations

a) Influence Line for R_A (Vertical Reaction at Support A)

- Equation:

$$R_A(x) = 1 - \frac{x}{8} \quad \text{for } 0 \leq x \leq 8 \text{ m}$$

- Key Coordinates:

- At load position B ($x = 0$ m): $R_A = 1 - \frac{0}{8} = +1.0$
- At load position C ($x = 4$ m): $R_A = 1 - \frac{4}{8} = +0.5$
- At load position D ($x = 8$ m): $R_A = 1 - \frac{8}{8} = 0.0$

b) Influence Line for V_C (Shear Force at Point C)

- Method of Sections at Point C ($x_C = 4$ m):

- Case 1: Load on segment BC ($0 \leq x < 4$ m):

$$V_C(x) = R_A - 1 = \left(1 - \frac{x}{8}\right) - 1 = -\frac{x}{8}$$

- At $x = 0$ m (Joint B): $V_C = 0.0$
- At $x \rightarrow 4^-$ m (Just left of C): $V_C = -\frac{4}{8} = -0.5$

- Case 2: Load on segment CD ($4 \text{ m} < x \leq 8 \text{ m}$):

$$V_C(x) = R_A = 1 - \frac{x}{8}$$

- At $x \rightarrow 4^+$ m (Just right of C): $V_C = 1 - \frac{4}{8} = +0.5$
- At $x = 8$ m (Joint D): $V_C = 1 - \frac{8}{8} = 0.0$

- Unit Jump at Point C :

$$\Delta V_C = V_C(4^+) - V_C(4^-) = +0.5 - (-0.5) = 1.0$$

c) Influence Line for M_C (Bending Moment at Point C)

- Sign Convention: Positive bending moment causes sagging (tension at the bottom fibers).
- Equations by Segment:

CHAPTER 01 INFLUENCE LINES OF DETERMINATE AND INDETERMINATE STRUCTURES

- Case 1: Load on segment BC ($0 \leq x \leq 4$ m):

$$M_C(x) = R_E \cdot 4 = \left(\frac{x}{8}\right) \cdot 4 = \frac{x}{2}$$

- At $x = 0$ m (Joint B): $M_C = 0.0$ kN · m
- At $x = 4$ m (Point C): $M_C = \frac{4}{2} = +2.0$ kN · m

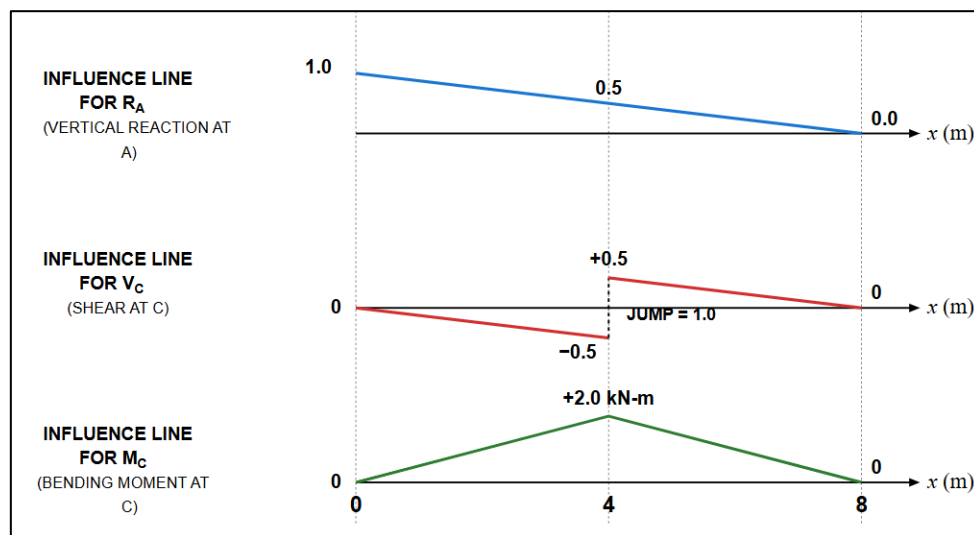
- Case 2: Load on segment CD ($4 \leq x \leq 8$ m):

$$M_C(x) = R_A \cdot 4 = \left(1 - \frac{x}{8}\right) \cdot 4 = 4 - \frac{x}{2}$$

- At $x = 4$ m (Point C): $M_C = 4 - \frac{4}{2} = +2.0$ kN · m
- At $x = 8$ m (Joint D): $M_C = 4 - \frac{8}{2} = 0.0$ kN · m

3. Summary Table of Coordinates (Influence Line Values)

Load Position (x)	Joint B (0 m)	Point C- (4- m)	Point C+ (4+ m)	Joint D (8 m)
R_A (Vertical Reaction at A)	+1.0	+0.5	+0.5	0.0
V_C (Shear Force at C)	0.0	-0.5	+0.5	0.0
M_C (Bending Moment at C)	0.0 kN · m	+2.0 kN · m	+2.0 kN · m	0.0 kN · m

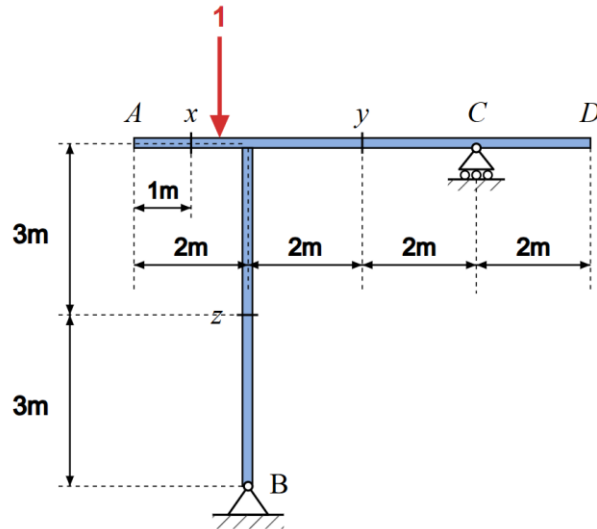


Example

Influence Lines for Frames

Example: Draw the I.L. for the following functions for the frame shown below.

R_B , R_C , V_{By} , V_y , V_x , V_z , A_z , M_y and M_x



Solution

Support Reactions (R_B and R_C)

To determine the influence lines for the support reactions, a unit vertical load $P = 1$ moves along the top beam AD ($0 \leq x_p \leq 8$ m).

- **Moment equilibrium about Support B ($\sum M_B = 0$):**

$$R_C \cdot 4 - 1 \cdot (x_p - 2) = 0 \Rightarrow R_C = \frac{x_p - 2}{4}$$

- **Vertical force equilibrium ($\sum F_y = 0$):**

$$R_B + R_C - 1 = 0 \Rightarrow R_B = \frac{6 - x_p}{4}$$

- **Horizontal reaction at Support B (H_B):**

$$H_B = 0 \quad (\text{for all } x_p)$$

Since no external horizontal forces act on the frame, the horizontal reaction remains zero.

2. Internal Forces at Section x ($x_p = 1$ m)

- **Shear Force (V_x):**
 - For $0 \leq x_p < 1$ m (load to the left of section x): $V_x = -1$
 - For $1 \text{ m} < x_p \leq 8$ m (load to the right of section x): $V_x = 0$
- **Bending Moment (M_x):**
 - For $0 \leq x_p \leq 1$ m: $M_x = x_p - 1$
 - For $1 \text{ m} \leq x_p \leq 8$ m: $M_x = 0$

3. Shear Force Just Right of Column Joint (V_{By})

This corresponds to the shear force in the horizontal beam segment immediately to the right of the joint above column B ($x_p = 2^+$):

- For $0 \leq x_p < 2$ m: $V_{By} = R_C = \frac{x_p - 2}{4}$
- For $2 \text{ m} < x_p \leq 8$ m: $V_{By} = R_C - 1 = \frac{x_p - 6}{4}$

4. Internal Forces at Section y ($x_p = 4$ m)

- **Shear Force (V_y):**

- For $0 \leq x_p < 4$ m: $V_y = R_C = \frac{x_p^{-2}}{4}$
- For $4 \text{ m} < x_p \leq 8$ m: $V_y = R_C - 1 = \frac{x_p^{-6}}{4}$

• **Bending Moment (M_y):**

- For $0 \leq x_p \leq 4$ m: $M_y = -2R_C = -\frac{x_p^{-2}}{2}$
- For $4 \text{ m} \leq x_p \leq 8$ m: $M_y = \frac{x_p^{-6}}{2}$

5. Internal Forces at Section z (Vertical Column JB)

• **Axial Force (A_z / Normal Force):**

The vertical column transmits the reaction R_B (compression when negative) :

$$A_z = -R_B = \frac{x_p - 6}{4}$$

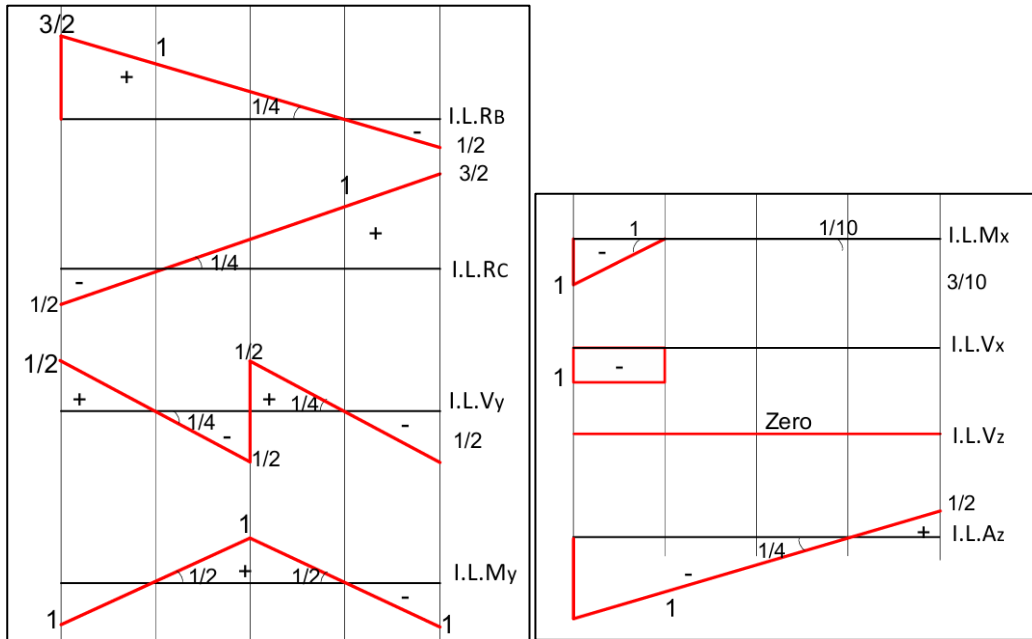
• **Column Shear Force (V_z):**

$$V_z = 0 \quad (\text{for all } x_p)$$

6. Summary Table of Influence Line Ordinates

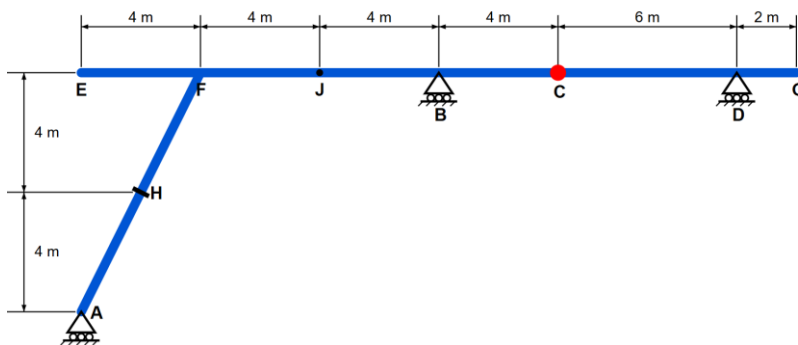
Function / Load Position	Point A (xp=0 m)	Section x (xp=1 m)	Joint B (xp=2 m)	Section y (xp=4 m)	Support C (xp=6 m)	Point D (xp=8 m)
R_C (Reaction at C)	-0.5	-0.25	0.0	+0.5	+1.0	+1.5
R_B (Reaction at B)	+1.5	+1.25	+1.0	+0.5	0.0	-0.5
V_x (Shear at x)	-1.0	-1.0 0.0	0.0	0.0	0.0	0.0
M_x (Moment at x)	-1.0	0.0	0.0	0.0	0.0	0.0
V_{By} (Shear right of B)	-0.5	-0.25	0.0/-1.0	-0.5	0.0	+0.5
V_y (Shear at y)	-0.5	-0.25	0.0	+0.5/-0.5	0.0	+0.5
M_y (Moment at y)	+1.0	+0.5	0.0	-1.0	0.0	+1.0
A_z (Axial Force in Column)	-1.5	-1.25	-1.0	-0.5	0.0	+0.5
V_z (Shear in Column)	0.0	0.0	0.0	0.0	0.0	0.0

CHAPTER 01 INFLUENCE LINES OF DETERMINATE AND INDETERMINATE STRUCTURES

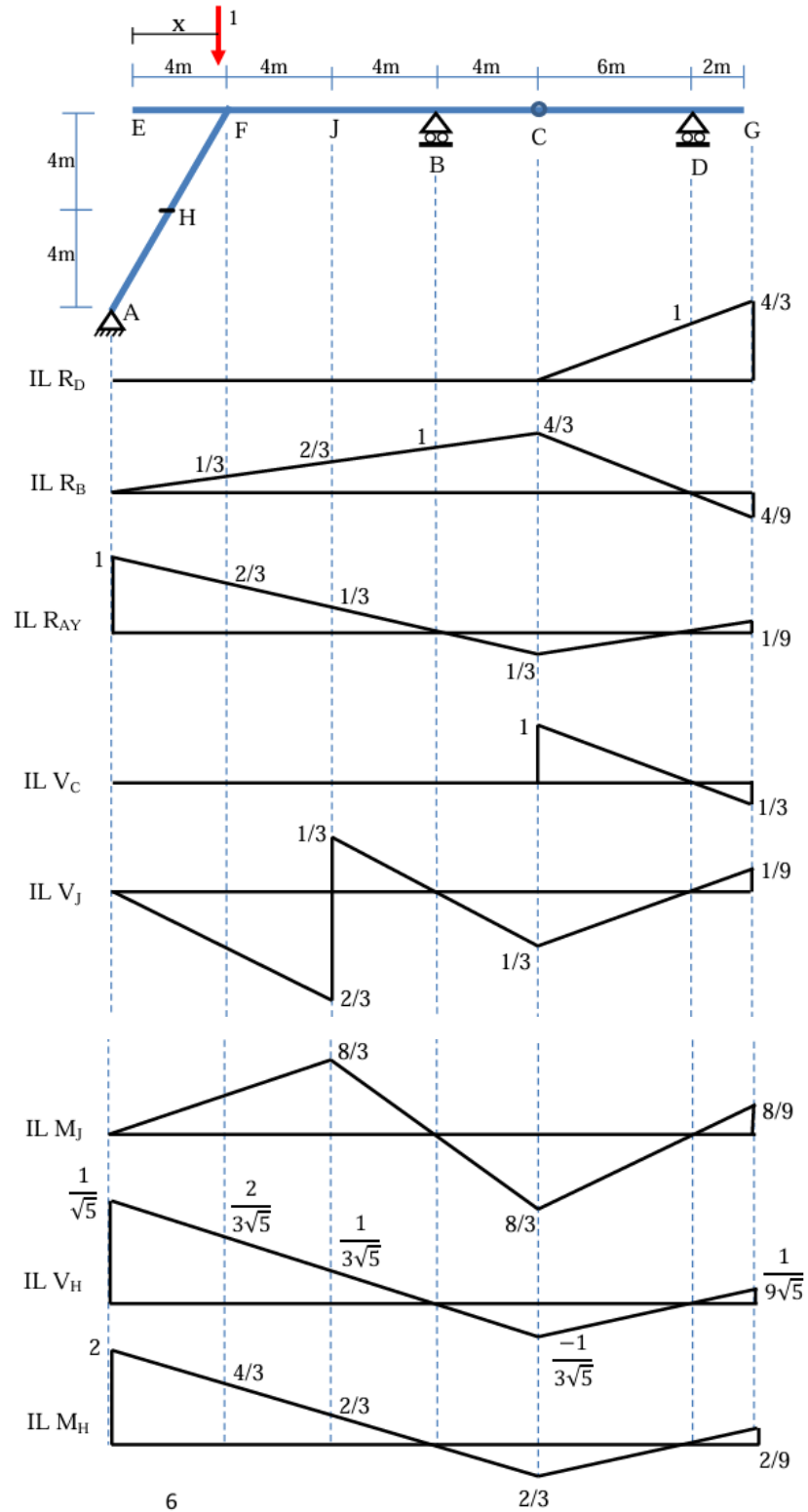


Example 03

Draw the I.L. for the following functions for the frame shown below. RB RC, RAY, Vc, VH, VJ, MH and MJ.



Solution



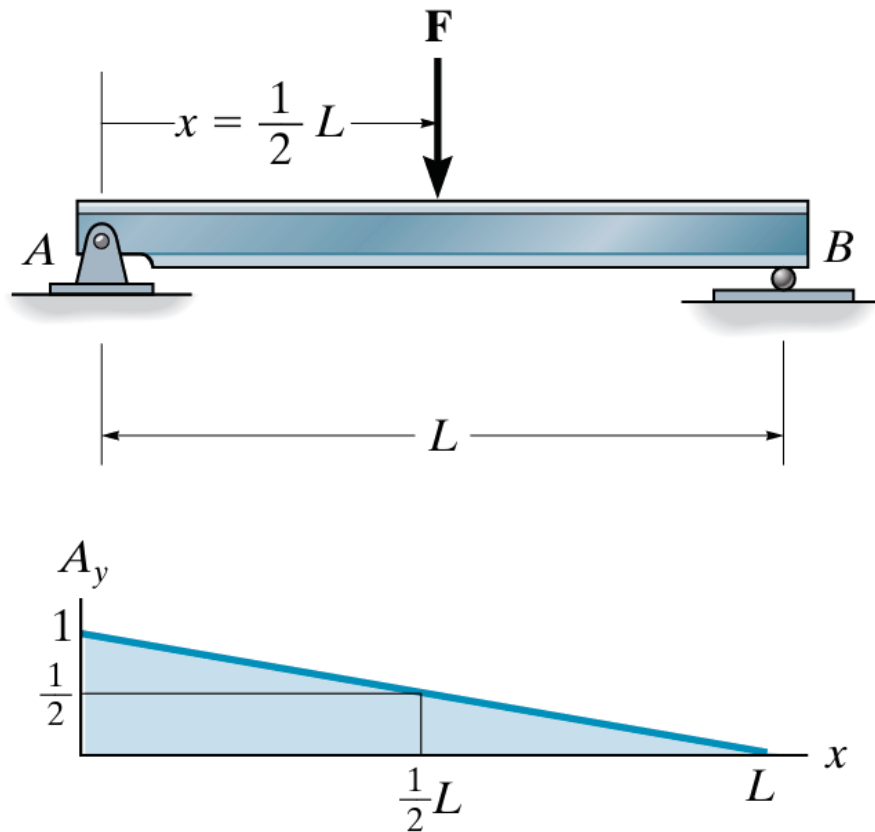
I.7 Application of Influence Lines :

Once the influence line for a function (reaction, shear, or moment) has been constructed, it will then be possible to position the live loads on the beam which will produce the maximum value of the function. Two types of loadings will now be considered.

I.7.1. Response at a particular location due to a single moving concentrated load

The numerical values of an **influence line** are determined using a **dimensionless unit load**. Therefore, for any concentrated force F acting on a beam at a position x , the value of the quantity being studied can be obtained by multiplying the ordinate of the influence line at that position by the magnitude of the force F .

For example, consider the **influence line for the reaction at A** for beam AB (Fig).



influence line for A_y

If the unit load is placed at:

$$x = \frac{1}{2}L$$

The influence line indicates that the reaction at A is:

$$A_y = \frac{1}{2}$$

Therefore, if a concentrated force F is applied at the same location, the reaction at A is:

$$A_y = \frac{1}{2}F$$

The same result can, of course, also be obtained directly using the **equations of statics**.

In general:

$$\text{Effect of } F = F \times \text{ordinate of the influence line}$$

The maximum effect produced by a concentrated force F occurs when the force is placed at the location where the **ordinate of the influence line is maximum**.

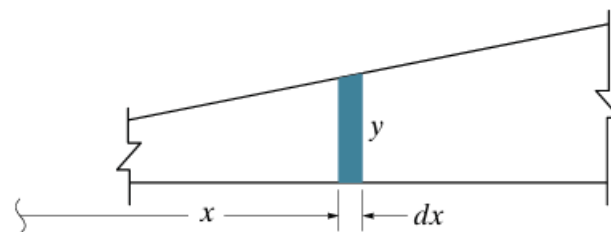
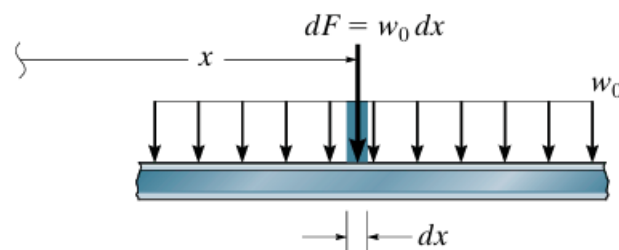
In this example, the maximum occurs at $x = 0$, at point A. The corresponding reaction is therefore:

$$A_y = (1)F = F$$

Key point: For a concentrated force, simply read the ordinate of the influence line at the force location and multiply it by the magnitude of F .

I.7.2. Response at a particular location due to a uniformly distributed live load

Consider a portion of a beam subjected to a **uniformly distributed load** w_0 , as shown in Fig.



influence line for function

Each small segment dx of the load can be considered as a **concentrated force**:

$$dF = w_0 dx$$

If dF is applied at a position x , where the ordinate of the influence line for a particular function (reaction, shear, or moment) is y , then the corresponding effect is:

$$dF(y) = (w_0 dx)y$$

The total effect of all the concentrated forces dF is obtained by integrating over the entire loaded length:

$$\int w_0 y dx = w_0 \int y dx$$

Since

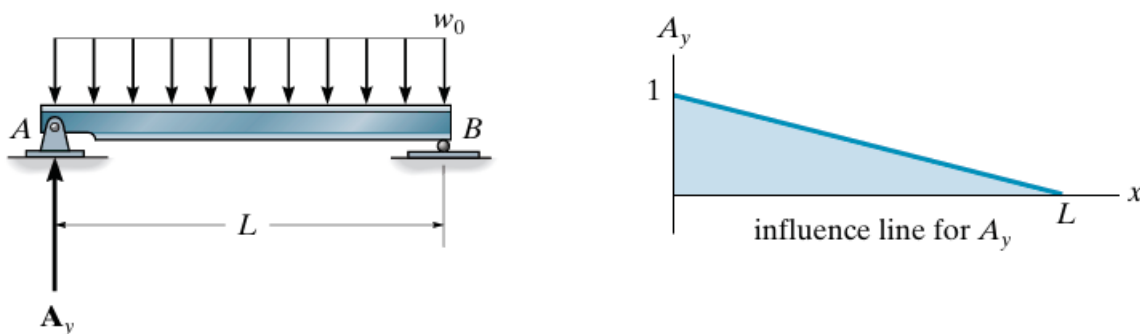
$$\int y dx$$

represents the **area under the influence line**, the general rule is:

$$\text{Effect of a uniform load} = (\text{area under the influence line}) \times w_0$$

Example

For the uniformly loaded beam shown in Fig., the reaction at A can be determined directly from the influence line.



The area under the influence line is a triangle with:

- Base = L
- Height = 1

Therefore:

$$A_y = (\text{Area})(w_0)$$

$$A_y = \left(\frac{1}{2}L \times 1\right)w_0$$

Hence:

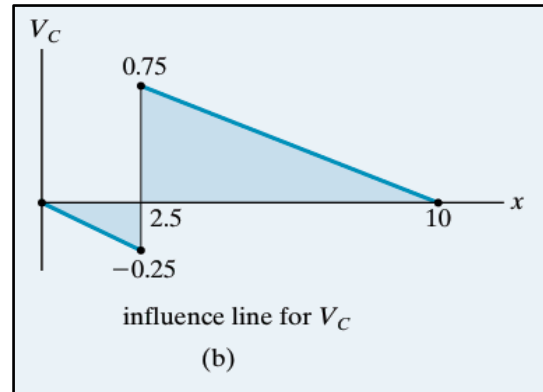
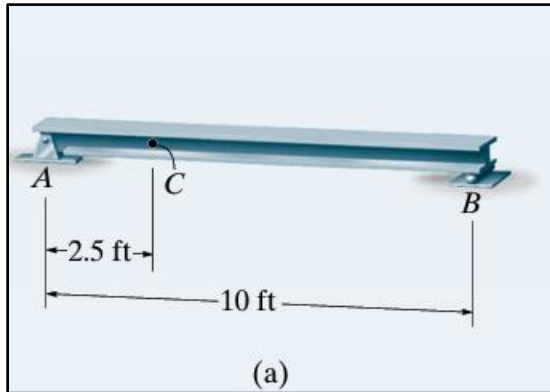
$$A_y = \frac{1}{2}w_0L$$

This result can, of course, also be obtained directly using the **equations of statics**.

Key point: For a uniformly distributed load, the effect is obtained by multiplying the **area under the influence line** by the **load intensity** w_0

Example :

Determine the maximum positive shear that can be developed at point C in the beam shown in **Fig.a** due to a concentrated moving load of 4000 lb and a uniform moving load of 2000 lb/ft.



SOLUTION The influence line for the shear at C has been established in Example 4 Fig.14 and is shown in Fig. b.

1. Concentrated Force.

The maximum positive shear at C will occur when the 4000-lb force is located at $x = 2.5$ ft, since this is the positive peak of the influence line. The ordinate of this peak is $+0.75$; so that $V_C = 0.75(4000 \text{ lb}) = 3000 \text{ lb}$

2. Uniform Load.

The uniform moving load creates the maximum positive influence for V_C when the load acts on the beam between $x = 2.5$ ft and $x = 10$ ft, since within this region the influence line has a positive area.

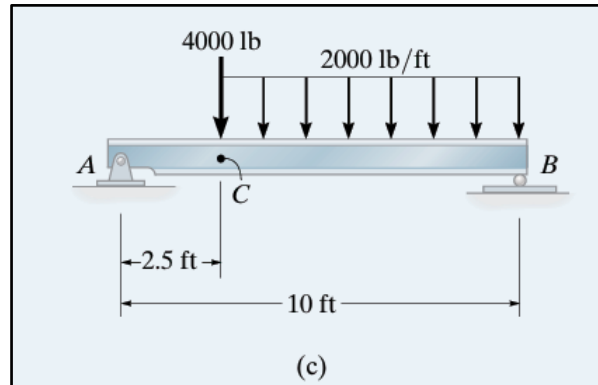
The magnitude of V_C due to this loading is

$$V_C = [1/2(10 - 2.5)*(0.75)]*2000 = 5625 \text{ lb}$$

Total Maximum Shear at C :

$$(V_C)_{\max} = 3000 \text{ lb} + 5625 \text{ lb} = 8625 \text{ lb}$$

Notice that once the positions of the loads have been established using the influence line, **Fig.c**, this value of $(V_C)_{\max}$ can also be determined using statics and the method of sections. Show that this is the case



Verification Using Statics

Once the critical positions of the loads have been determined from the influence line :

- The 4000-lb concentrated force is placed at $x = 2.5^+$ ft.
- The 2000-lb/ft uniform load acts from $x = 2.5$ ft to $x = 10$ ft.

1. Replace the uniform load by its resultant

The length over which the uniform load acts is

$$L = 10 - 2.5 = 7.5 \text{ ft}$$

Therefore, its equivalent concentrated force is

$$W = wL$$

$$W = (2000)(7.5) = 15,000 \text{ lb}$$

This resultant acts at the center of the loaded length:

$$x_R = 2.5 + \frac{7.5}{2} = 6.25 \text{ ft}$$

2. Determine the reaction at A

Taking moments about support B:

$$\sum M_B = 0$$

$$A_y(10) = 4000(10 - 2.5) + 15000(10 - 6.25)$$

Thus,

$$A_y = \frac{4000(7.5) + 15000(3.75)}{10}$$

$$A_y = \frac{30,000 + 56,250}{10}$$

$$A_y = 8625 \text{ lb}$$

3. Determine the shear at C

Now make a section immediately to the right of C, at C^+ .

Since both the 4000-lb force and the distributed load are located **to the right of C** , they do not act on the left segment.

For the left segment:

$$\sum F_y = 0$$

$$A_y - V_C = 0$$

Therefore,

$$V_C = A_y$$

$$\boxed{V_C = 8625 \text{ lb}}$$

Final Answer

Hence, using statics and the method of sections,

$$\boxed{(V_C)_{\max} = 8625 \text{ lb}}$$

which agrees exactly with the result obtained from the influence-line method:

$$\boxed{(V_C)_{\max} = 3000 + 5625 = 8625 \text{ lb}}$$

Thus, **the influence line determines the critical load positions, while statics verifies the resulting maximum shear at C .**

I.7.3. Response at a particular location due to a series of moving concentrated loads

When a train or truck moves across a bridge, several concentrated **wheel loads** act on the girder. To determine the **maximum reaction, shear, or moment** at a particular point, the influence line can be used.

For a **single concentrated load**, the maximum effect is simply:

$$\text{Maximum effect} = P \times (\text{maximum influence-line ordinate})$$

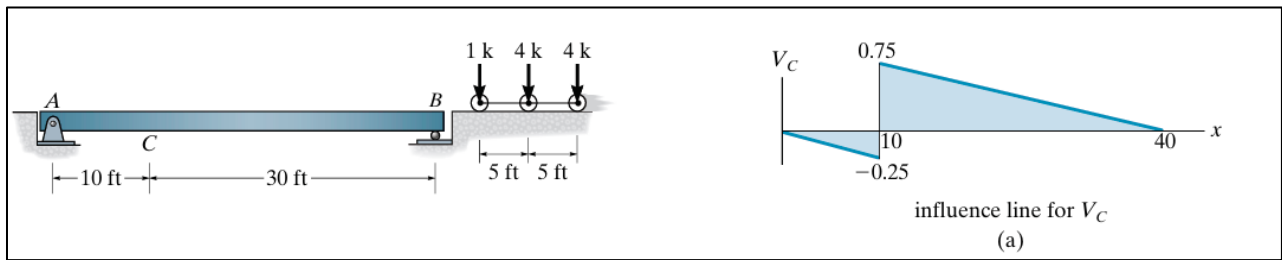
However, when **several concentrated loads move together**, the loads have different positions relative to the influence line. Therefore, the maximum effect must be determined by considering the possible critical positions of the load system.

Two approaches can be used:

1. **Trial-and-error method**
2. A method based on the **change in the reaction, shear, or moment** as the loads move.

Example: Maximum Shear at C

Consider the simply supported beam shown in Fig., with the influence line for the shear at C.



We want to determine the **maximum positive shear at C** caused by three moving concentrated loads: **1 k, 4 k, 4 k**

The loads move from **right to left**.

The positive peak of the influence line occurs immediately to the **right of C**, where the ordinate is:

$$y = +0.75$$

Therefore, the critical position will occur when one of the loads is located just to the right of C. Because there are three loads, three possible cases can be investigated.

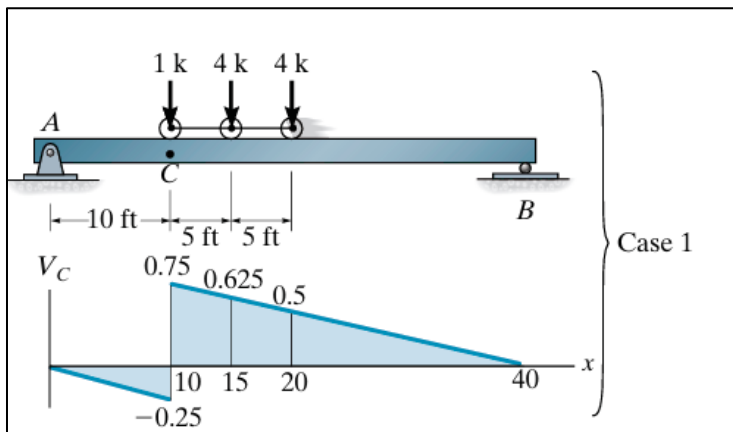
Case 1

The 1-k load is positioned at the positive peak:

$$(V_C)_1 = 1(0.75) + 4(0.625) + 4(0.50)$$

$$(V_C)_1 = 0.75 + 2.50 + 2.00$$

$$(V_C)_1 = 5.25 \text{ k}$$



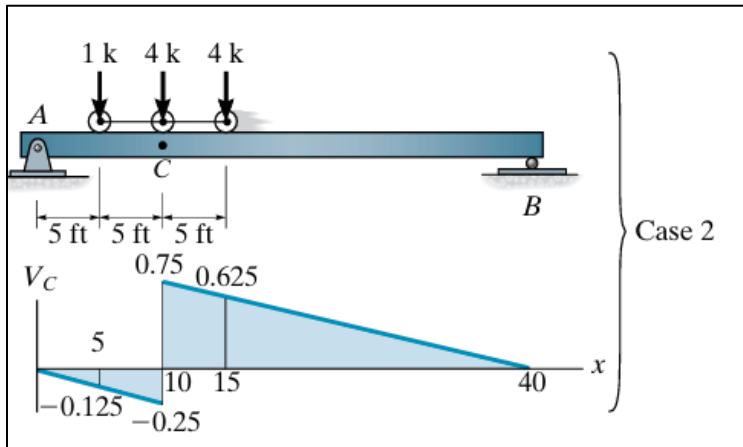
Case 2

The second 4-k load is positioned at the positive peak:

$$(V_C)_2 = 1(-0.125) + 4(0.75) + 4(0.625)$$

$$(V_C)_2 = -0.125 + 3.00 + 2.50$$

$$(V_C)_2 = 5.375 \text{ k}$$



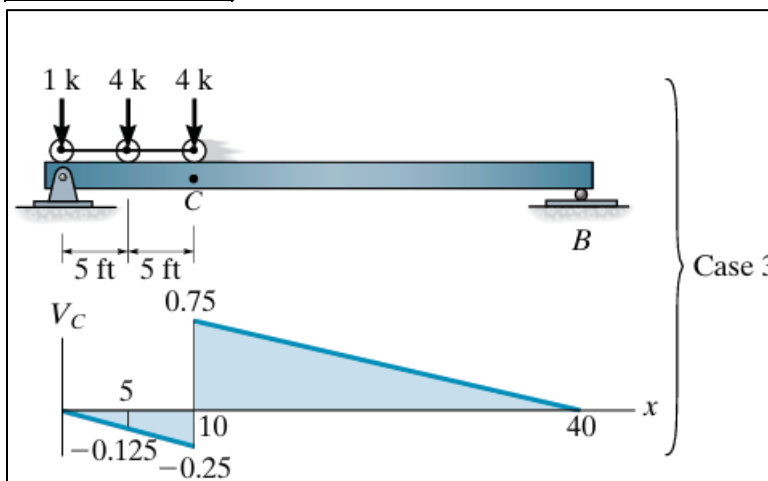
Case 3

The third 4-k load is positioned at the positive peak:

$$(V_C)_3 = 1(0) + 4(-0.125) + 4(0.75)$$

$$(V_C)_3 = 0 - 0.50 + 3.00$$

$$(V_C)_3 = 2.50 \text{ k}$$



Comparison

The three possible values are:

$$(V_C)_1 = 5.25 \text{ k}$$

$$(V_C)_2 = 5.375 \text{ k}$$

$$(V_C)_3 = 2.50 \text{ k}$$

Therefore,

$$(V_C)_{\max} = 5.375 \text{ k}$$

The **critical loading** is **Case 2**, where the 1-k load is located **5+ ft from the left support**, and the second 4-k load is immediately to the right of *C*.

Important idea

The key principle is:

For a series of concentrated moving loads, the maximum effect occurs when one of the loads is positioned at a critical point of the influence line.

We therefore place each possible load at the critical position, calculate the resulting effect, and compare the values.

Here:

Case 2 gives the maximum positive shear

because

$$5.375 > 5.25 > 2.50$$

3. A method based on the change in the reaction, shear, or moment as the loads move

Moving from one position to the next

The idea is to calculate the change in shear as the loads move from one position to the next:

$$\Delta V = \text{effect of the jumps} + \text{effect of moving along the slope}$$

In this example:

$$\Delta V = P_{\text{jump}}(\Delta y_{\text{jump}}) + \left(\sum P\right) s d$$

where:

- P_{jump} = the load that crosses the point where there is a jump;
- Δy_{jump} = the magnitude of the jump, including its sign;
- $\sum P$ = the sum of all the loads;
- s = the slope of the influence line;
- d = the distance the loads move.

The most important rule

$$\begin{array}{ll} \Delta V > 0 & \Rightarrow \text{continue moving the loads} \\ \Delta V < 0 & \Rightarrow \text{stop: the previous position is critical} \end{array}$$

This is precisely what the textbook means by “finding the change in shear as the loads move.

First Step: Calculate the Slope of the Influence Line

The slope of the influence line is:

$$s = \frac{0.75}{40 - 10}$$

Therefore,

$$s = \frac{0.75}{30}$$

$$\boxed{s = 0.025}$$

This means that when the loads move 1 ft along this inclined portion of the influence line, the ordinate increases by:

$$0.025$$

For a displacement of 5ft:

$$\Delta y = (0.025)(5) = 0.125$$

4. What Happens Between Case 1 and Case 2?

The loads move:

$$5 \text{ ft}$$

to the left.

During this movement:

Effect 1: Movement Along the Slope

All three loads move upward along the inclined portion of the influence line.

The sum of the loads is:

$$1 + 4 + 4 = 9 \text{ k}$$

The increase in the ordinate is:

$$(0.025)(5) = 0.125$$

Therefore, the contribution due to the slope is:

$$9(0.125) = 1.125 \text{ k}$$

Effect 2: Jump at Point C

The 1 k load crosses point C.

At C, the influence line has a jump of:

$$+0.75 - (-0.25) = 1$$

However, during the movement considered, the load makes a downward jump on the influence-line diagram.

Therefore, its contribution is:

$$1(-1) = -1 \text{ k}$$

5. Total Change

We add the two effects:

$$V_{1-2} = 1(-1) + (1 + 4 + 4)(0.025)(5)$$

$$V_{1-2} = -1 + 9(0.125)$$

$$V_{1-2} = -1 + 1.125$$

$$\boxed{V_{1-2} = +0.125 \text{ k}}$$

The result is positive.

Therefore:

$$\boxed{V_2 > V_1}$$

In other words, Case 2 gives a larger shear than Case 1.

We can also verify this directly:

$$5.375 - 5.25 = 0.125 \text{ k}$$

which gives exactly the same result.

6. Now Consider Case 2 → Case 3

The loads move another:

$$5 \text{ ft}$$

This time, the 4 k load crosses point C.

Therefore, it experiences the negative jump:

$$4(-1) = -4 \text{ k}$$

At the same time, all three loads continue to move upward along the inclined portion of the influence line.

The contribution from the slope remains:

$$\begin{aligned} & (1 + 4 + 4)(0.025)(5) \\ & = 9(0.125) \\ & = 1.125 \text{ k} \end{aligned}$$

Therefore:

$$V_{2-3} = 4(-1) + (1 + 4 + 4)(0.025)(5)$$

$$V_{2-3} = -4 + 1.125$$

$$\boxed{V_{2-3} = -2.875 \text{ k}}$$

This time, the change is negative.

7. What Does the Negative Result Mean?

We found:

$$V_{1-2} = +0.125 \text{ k}$$

Therefore:

$$V_2 > V_1$$

But:

$$V_{2-3} = -2.875 \text{ k}$$

Therefore:

$$V_3 < V_2$$

So the shear increases from Case 1 to Case 2, but then decreases from Case 2 to Case 3.

Hence, the critical position is:

$$\boxed{\text{Case 2}}$$

This illustrates the key idea: as soon as ΔV changes from positive to negative, the preceding position is the critical position for maximum shear.

The Difference in One Mental Picture

Shear V_c

There is: a jump + a slope

$$\Delta V = P_{\text{jump}} \Delta y_{\text{jump}} + \sum P s d$$

Moment M_c

There is: no jump, but there may be two different slopes.

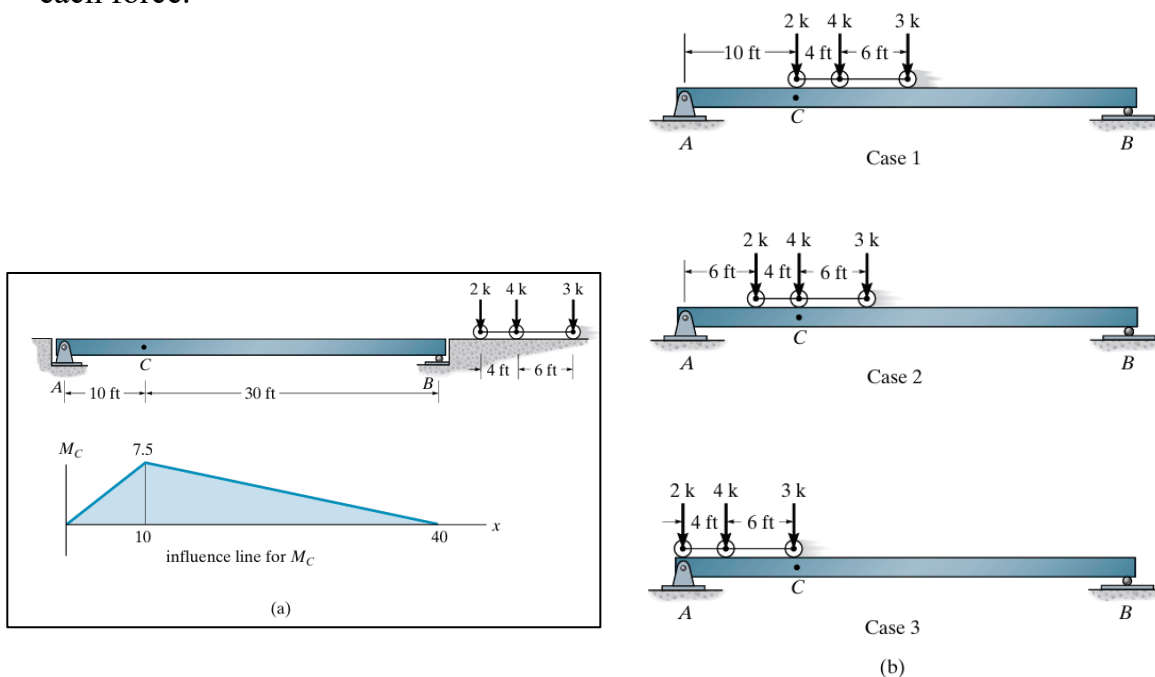
$$\Delta M = \sum P_i s_i d$$

In short

- Shear: jump + slope
- Moment: slopes only
- For moment, if different loads are on different slopes, calculate each group separately.

Example Moment.

We can also use the foregoing methods to determine the critical position of a series of concentrated forces so that they create the largest internal moment at a specific point in a structure. Of course, it is first necessary to draw the influence line for the moment at the point. As an example, consider the beam, loading, and influence line for the moment at point C shown in **Fig.** If each of the three concentrated forces is placed on the beam, coincident with the peak of the influence line, we will obtain the greatest influence from each force.



This is the same change in-effect method, but now it is used to find the maximum moment at C instead of shear.

The loads are moved from one case to the next, and we calculate only the change in moment.

1. From Case 1 to Case 2

The change in moment is:

$$M_{1-2} = -2 \left(\frac{7.5}{10} \right) (4) + (4 + 3) \left(\frac{7.5}{40 - 10} \right) (4)$$

$$M_{1-2} = 1.0 \text{ k} \cdot \text{ft}$$

Since:

$$M_{1-2} > 0$$

the moment increases when the loads move from Case 1 to Case 2.

Therefore, we must continue moving the loads.

2. From Case 2 to Case 3

The loads are moved another 6ft.

$$M_{2-3} = -(2 + 4) \left(\frac{7.5}{10} \right) (6) + 3 \left(\frac{7.5}{40 - 10} \right) (6)$$

$$M_{2-3} = -22.5 \text{ k} \cdot \text{ft}$$

Now the change is negative:

$$M_{2-3} < 0$$

Therefore, the moment decreases when we move from Case 2 to Case 3.

So the previous position, **Case 2**, is the critical loading position.

3. Calculate the maximum moment

Once Case 2 has been identified as the critical position, calculate the actual moment at *C*:

$$(M_C)_{\max} = 2(4.5) + 4(7.5) + 3(6.0)$$

$$= 9 + 30 + 18$$

$$(M_C)_{\max} = 57.0 \text{ k} \cdot \text{ft}$$

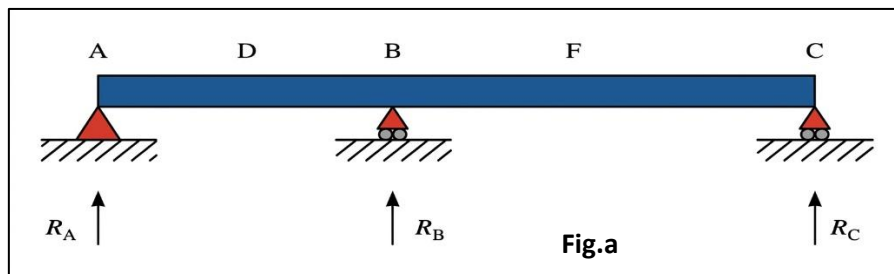
II. Influence Lines of Indeterminate Structures

The procedures used to determine influence lines for statically indeterminate structures are similar to those described for statically determinate structures.

- The main distinguishing characteristic between influence lines in the two cases is that, for statically determinate structures, the diagrams consist of straight-line segments, whereas for statically indeterminate structures, they are generally composed of curves.
- The **Müller-Breslau principle** makes it possible to directly obtain the **qualitative shape** of the influence-line diagrams.
- To determine the numerical values of the ordinates of the deformed curve representing the required influence line, the Müller-Breslau method must be complemented by another method, such as the **force method** or another appropriate structural analysis method.

II.1 Influence Lines for a Two-Span Continuous Beam Using the Müller-Breslau Principle

Consider the two-span continuous beam ABC shown in Fig. We will construct the influence lines for the reaction R_B , the shear force V_D , and the bending moment M_F using the **Müller-Breslau principle**.

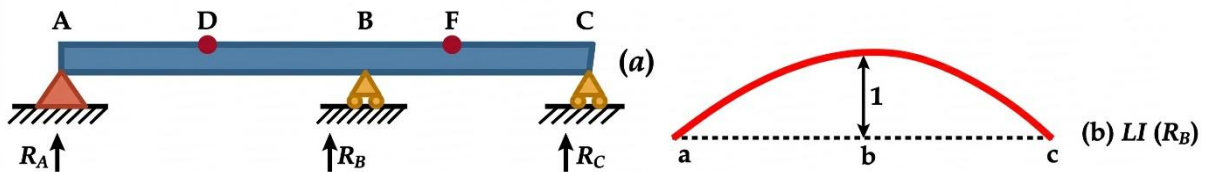


1. Influence Line for R_B

As shown in Fig.b:

- Remove the support (constraint) at B .
- Apply a **unit displacement** at B in the direction of the reaction R_B .
- The beam deforms while remaining supported at A and C .

The resulting **deformed shape of the beam** represents the **qualitative influence line** for R_B . Because the beam is **statically indeterminate**, the influence line is **curved**, rather than consisting of straight-line segments.



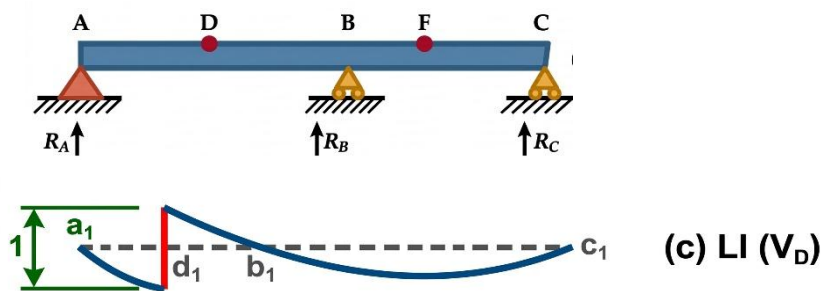
2. Influence Line for the Shear Force V_D

To obtain the influence line for the shear at D :

- Make a cut in the beam at D .
- Apply a **unit relative vertical displacement** between the two sides of the cut, as shown in Fig (c).
- The beam remains connected to the support at C , so its deformation is a **curve**.

The slope of the influence line must be the same on both sides of D . Otherwise, a residual moment would produce a relative rotation, which is not allowed because only a displacement corresponding to the shear force has been imposed.

Therefore, the influence line for V_D between A and D is **curved**.

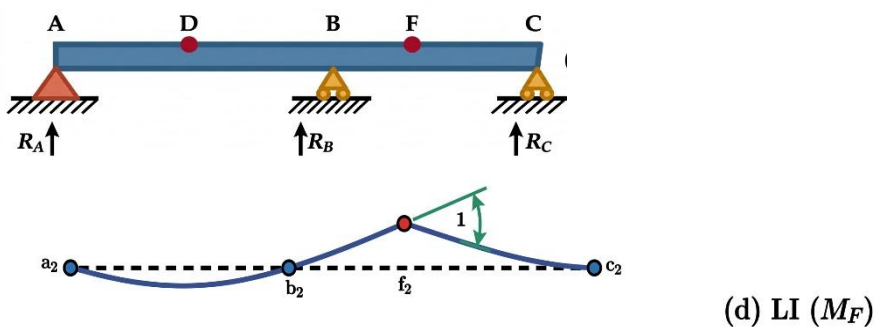


3. Influence Line for the Bending Moment M_F

To obtain the influence line for the bending moment at F :

- Insert a **hinge (internal pin)** at F .
- Apply a **unit relative rotation** between the two sides of the hinge, as shown in Fig.d.

According to the Müller-Breslau principle, the resulting deformed shape gives the qualitative influence line for M_F .



The influence line is therefore **curved** along both portions: ABF and FC

Key idea

For a **statically indeterminate continuous beam**, the Müller-Breslau principle gives the **qualitative shape** of the influence line by releasing the corresponding restraint and imposing the associated unit displacement:

Quantity	Release / Modification	Unit action
R_B	Remove support at B	Unit vertical displacement
V_D	Cut beam at D	Unit relative vertical displacement
M_F	Insert hinge at F	Unit relative rotation

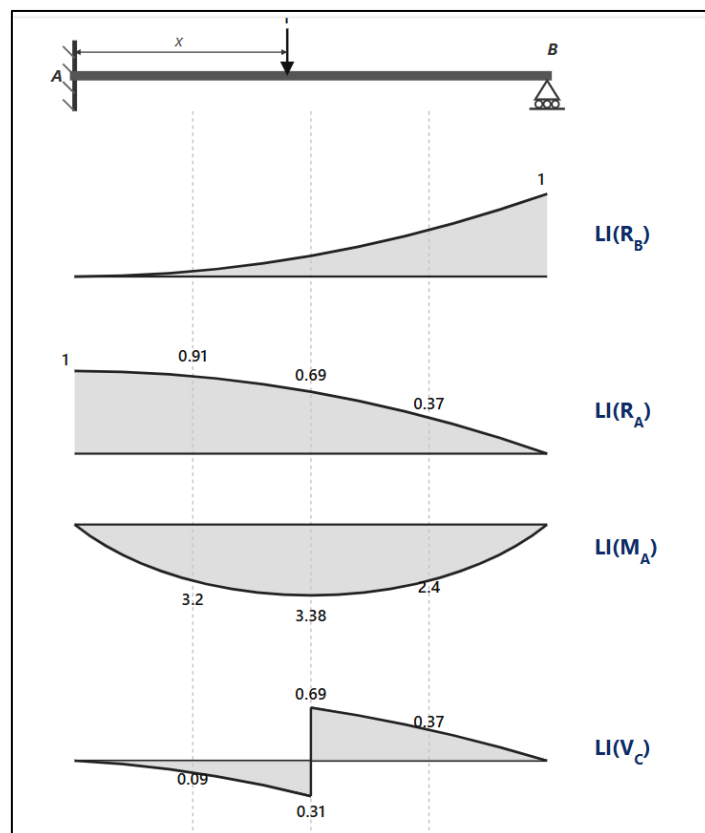
The important difference from a statically determinate beam is that the resulting influence lines are generally **curved**, because the continuous beam is **statically indeterminate**.

Example 01

Using the **Müller-Breslau principle**, sketch qualitatively the following influence lines:

- Influence line for : **IL(R_A)**
- Influence line for : **IL(R_B)**
- Influence line for : **IL(V_C)**

Solution



To obtain the numerical values shown on the influence lines, you need to determine the analytical equation for each internal force or reaction as a function of the unit load's position x ($P = 1$). (Cliquer sur lien pour bien comprendre

<https://www.youtube.com/watch?v=BEGfz0qrKzY>)

Let $a = \frac{x}{L}$ be the relative position of the unit load along the beam ($0 \leq a \leq 1$).

Step 1: Derivation of $R_B(a)$ using the Force Method

For a propped cantilever beam (fixed at A , simply supported at B — statically indeterminate of degree 1):

1. **Primary System** : Remove the support at B (converting the structure into a cantilever beam fixed at A).
2. **Deflection due to unit load**: The downward deflection at point B caused by a unit load $P = 1$ at position $x = aL$ is:

$$\delta_{BP} = \frac{L^3}{6EI} \cdot a^2(3 - a)$$

3. **Flexibility coefficient**: The upward deflection at B due to a unit reaction $R_B = 1$ is:

$$\delta_{BB} = \frac{L^3}{3EI}$$

4. **Compatibility Condition** ($\Delta_B = 0$):

$$\delta_{BP} - R_B \cdot \delta_{BB} = 0 \Rightarrow R_B(a) = \frac{\delta_{BP}}{\delta_{BB}} = \frac{3a^2 - a^3}{2}$$

Step 2: Numerical Values for Each Influence Line

The values on the diagrams are evaluated at quarter points of the span ($a = 0.25$, $a = 0.50$, $a = 0.75$):

1. Influence Line for R_A : $IL(R_A)$

From vertical equilibrium ($\sum F_y = 0 \Rightarrow R_A + R_B = 1$):

$$R_A(a) = 1 - R_B(a) = 1 - \frac{3a^2 - a^3}{2} = \frac{(1 - a)^2(2 + a)}{2}$$

- At $a = 0$ ($x = 0$): $R_A = 1$
- At $a = 0.25$ ($x = 0.25L$): $R_A = \frac{(0.75)^2 \times 2.25}{2} \approx \mathbf{0.91}$
- At $a = 0.50$ ($x = 0.50L$): $R_A = \frac{(0.50)^2 \times 2.50}{2} = \mathbf{0.69}$
- At $a = 0.75$ ($x = 0.75L$): $R_A = \frac{(0.25)^2 \times 2.75}{2} \approx \mathbf{0.37}$

- At $a = 1.00$ ($x = L$): $R_A = 0$

2. Influence Line for Shear Force at C: $IL(V_C)$ (where C is at mid-span, $a_C = 0.5$)

By equilibrium of the section to the left or right of C:

- **When the load is to the left of C ($a \leq 0.5$):** $V_C(a) = -R_B(a)$
 - At $a = 0.25$: $V_C = -\frac{3(0.25)^2 - (0.25)^3}{2} \approx -0.09$
 - At $a = 0.50^-$: $V_C = -\frac{3(0.5)^2 - (0.5)^3}{2} = -0.31$
- **When the load is to the right of C ($a > 0.5$):** $V_C(a) = +R_A(a)$
 - At $a = 0.50^+$: $V_C = +0.69$
 - At $a = 0.75$: $V_C = +0.37$

The jump at section C is exactly $\Delta V_C = 0.69 - (-0.31) = 1.00$.

3. Influence Line for Bending Moment at A: $IL(M_A)$

From moment equilibrium about A:

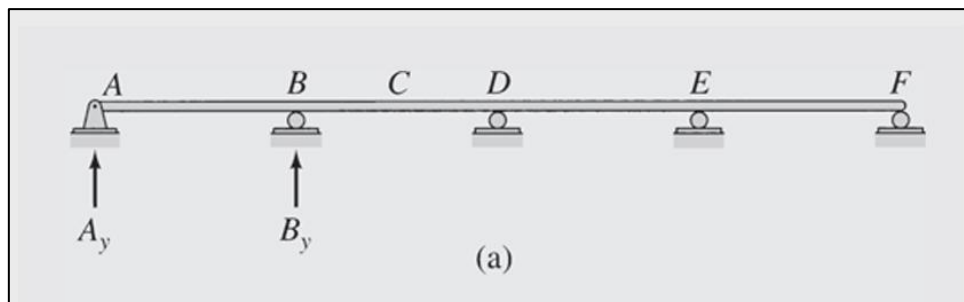
$$M_A(a) = R_B(a) \cdot L - a \cdot L = -\frac{L \cdot a(1-a)(2-a)}{2}$$

For a given span length (for instance $L = 18$ m) :

- At $a = 0.25$: $|M_A| \approx 3.2$
- At $a = 0.50$: $|M_A| = \frac{18 \times 0.5 \times 0.5 \times 1.5}{2} = 3.38$
- At $a = 0.75$: $|M_A| \approx 2.4$

Example 02

Draw qualitative influence lines for the vertical reactions at supports A and B, the bending moment at point B, and the shear and bending moment at point C of the four-span continuous beam shown in Fig (a). Also, show the arrangements of a uniformly distributed downward live load w_l to cause the maximum positive reactions at supports A and B, the maximum negative bending moment at B, the maximum negative shear at C, and the maximum positive bending moment at C.



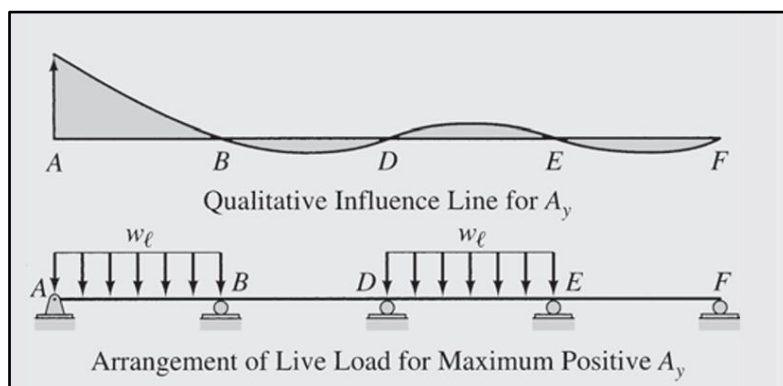
Solution

For the **four-span continuous beam**, the Müller-Breslau principle is used.

1. Influence line for A_y

- Remove the vertical support at **A**.
- Give A a small displacement in the positive direction of A_y .
- The deflected shape gives the **qualitative influence line for A_y** .
- The maximum positive A_y occurs when the load w_l covers the spans where the influence line is **positive**:

AB and DE

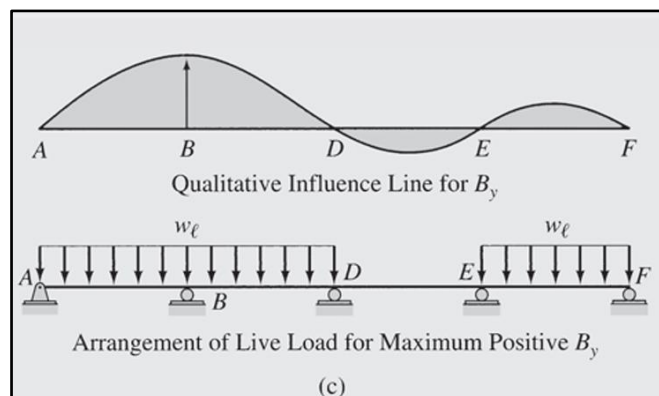


2. Influence line for B_y

- Remove the vertical restraint at **B**.
- Give B a small displacement in the positive direction of B_y .
- The resulting deflected shape gives the influence line for B_y .

To obtain maximum positive B_y , place w_l on the spans where the influence line is **positive**.

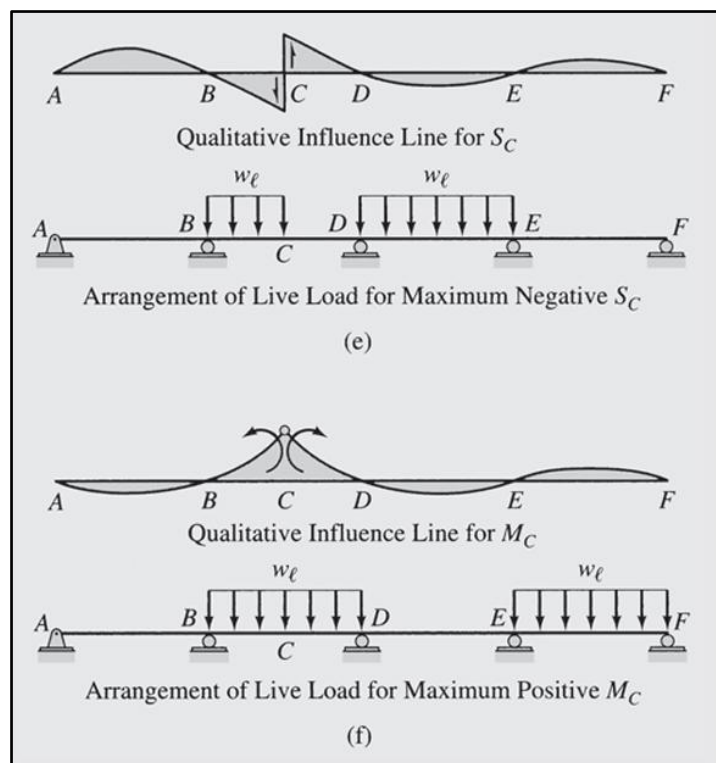
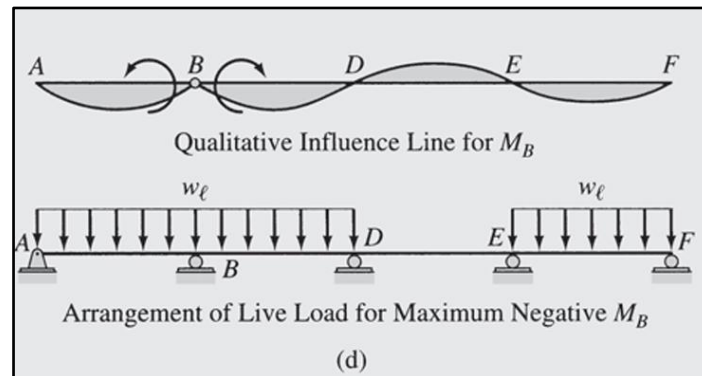
AD and EF



3. Influence line for M_B

- Insert a **hinge** at **B**.
- Rotate the left part counterclockwise and the right part clockwise.
- The resulting deflected shape gives the influence line for M_B .
- For the **maximum negative moment at B**, place w_l where the influence line is **negative**:

AB, BD, EF



Key rule

Maximum positive effect $\rightarrow$ load positive ordinates

Maximum negative effect $\rightarrow$ load negative ordinates

I.8. Problems